

# Division-of-Thoughts: Harnessing Hybrid Language Model Synergy for Efficient On-Device Agents

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## Abstract

The rapid expansion of web content has made on-device AI assistants indispensable for helping users manage the increasing complexity of online tasks. The emergent reasoning ability in large language models offer a promising path for next-generation on-device AI agents. However, deploying full-scale Large Language Models (LLMs) on resource-limited local devices is challenging. In this paper, we propose **Division-of-Thoughts (DoT)**, a collaborative reasoning framework leveraging the synergy between locally deployed Smaller-scale Language Models (SLMs) and cloud-based LLMs. DoT leverages a *Task Decomposer* to elicit the inherent planning abilities in language models to decompose user queries into smaller sub-tasks, which allows hybrid language models to fully exploit their respective strengths. Besides, DoT employs a *Task Scheduler* to analyze the pair-wise dependency of sub-tasks and create a dependency graph, facilitating parallel reasoning of sub-tasks and the identification of key steps. To allocate the appropriate model based on the difficulty of sub-tasks, DoT leverages a *Plug-and-Play Adapter*, which is an additional task head attached to the SLM that does not alter the SLM’s parameters. To boost adapter’s task allocation capability, we propose a self-reinforced training method that relies solely on task execution feedback. Extensive experiments on various benchmarks demonstrate that our DoT significantly reduces LLM costs while maintaining competitive reasoning accuracy. Specifically, DoT reduces the average reasoning time and API costs by 66.12% and 83.57%, while achieving comparable reasoning accuracy with the best baseline methods.<sup>1</sup>

## CCS Concepts

- **Computing methodologies** → **Natural language generation**;
- **Computer systems organization** → *Client-server architectures*.

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<sup>1</sup>Code available at: <https://github.com/tsinghua-fib-lab/DoT>



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# 思想划分：利用混合语言模型协同作用实现高效的设备端代理

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## 关键词

大语言模型，LLM 推理，人工智能代理，边缘 - 云协作

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## 1 引言

随着网络内容的指数增长，设备上的人工智能助手已成为帮助用户应对日益复杂的在线环境的重要工具。这一趋势促使个人助手广泛普及，例如谷歌助手、苹果 Siri、亚马逊 Alexa、阿里巴巴天猫精灵和小米小 AI，它们已证明在帮助用户处理海量网络内容方面非常有效，如网页浏览、内容搜索、在线购物和旅行计划。这些由人工智能驱动的助手使网络应用能够利用人工智能技术的快速进步，提供更个性化和便捷的用户体验。在近年来大语言模型 (LLM) 的人工智能突破中，常识推理和上下文学习的新兴能力被广泛认为是下一代设备端助手的关键组成部分。因此，革命性地改变人工智能个人助理

然而，在本地设备上部署大型语言模型（LLM）代理面临重大挑战，因为在资源受限的设备（如智能手机和个人电脑）上运行拥有数万亿参数的 LLM 是不现实的 [41]。相反，仅依赖于云的商业 LLM 会引发隐私风险、不可靠的连接以及高昂的经济成本的担忧 [22]。近期的研究集中在训练小规模语言模型和开发模型压缩技术 [23, 24, 40]，目标是创建可在本地设备上实际部署的参数低于 100 亿的模型，例如 Llama 3 系列 [10] 和 Phi 3 系列 [1]。然而，这种方法在训练或压缩这些模型时引入了额外的计算成本，并不可避免地导致

## CCS 概念

- **计算方法** → **自然语言生成**；
- **计算机系统组织** → **客户端 - 服务器架构**。

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performance degradation compared to full-size LLMs. Preliminary efforts have also explored the potential of edge-cloud collaboration, where tasks exceeding the capabilities of locally deployed Smaller-scale Language Models (SLMs) are rerouted to more powerful cloud-based LLMs [5, 13]. Despite this, the significant performance gap between SLMs and LLMs often leads to a suboptimal trade-off between reasoning capabilities and cost.

To address this problem, our work draws inspiration from the fundamental economic concept of “division of labour”, which posits that breaking down complex tasks into finer components often leads to more efficient solutions by allowing collaborative partners to fully exploit their respective strengths. Building on this idea, we propose a novel Division-of-Thoughts (**DoT**) framework to fully harness the synergy between locally deployed SLMs and cloud-based LLMs through sophisticated task decomposition and optimized sub-tasks allocation. Specifically, DoT leverages a *Task Decomposer*, a meta-prompt that combines “chain-of-thought”-like prompting [38] with carefully curated task decomposition examples. This approach taps into the inherent planning abilities of language models [35], enabling them to decompose user queries into smaller sub-tasks. Our core insight is that even complex user queries often contain a significant portion of simple sub-tasks that can be adequately handled by SLMs. Therefore, decomposing user queries can lead to optimized collaboration on sub-task level, i.e., unleashing the potential of “Division-of-Thoughts”. Moreover, DoT employs a *Task Scheduler* that analyzes the pair-wise dependency between sub-tasks and creates a dependency graph. This facilitates efficient and accurate scheduling by identifying parallel sub-tasks and assessing the structural importance of each task. More importantly, we design a self-reinforced training method to boost the task allocation capability of SLM, requiring no human annotation, but only the feedback of task execution. Our training method leverages a novel tree search algorithm that accounts for both the uncertainty of language models and the task performance to optimize sub-task allocation decisions. As a result, we can create a high-quality sub-task allocation dataset without human supervision, based on which, we train a lightweight adapter that significantly boosts the SLM’s ability to allocate sub-tasks. It is important to note that the adapter fully preserves the general capabilities of the SLM, as it does not modify SLM’s parameters. Instead, it introduces a detachable decoding head specialized for sub-task allocation.

We conduct extensive experiments on seven widely adopted LLM agent benchmarks, covering a variety of scenarios, including logical reasoning, web browsing, math problems solving, and commonsense reasoning. The results demonstrate that our DoT framework significantly reduces LLM costs while maintaining reasoning accuracy comparable to the best baseline methods across all benchmarks. Specifically, the average reasoning time and API costs are reduced by 66.12% and 83.57%, respectively, compared to the most accurate baselines. Additionally, an ablation study confirms the effectiveness of our key model design choices. Moreover, the DoT framework consistently achieves superior cost-accuracy trade-offs compared to task referral methods that do not leverage “Division-and-Allocate” strategy, where the performance gain are particularly large at low budget setting. We further demonstrate that DoT exhibits strong generalization across benchmarks.

To summarize, our contributions are three-fold:

- We present a novel Division-of-Thoughts (DoT) framework that fully exploits the synergy between locally deployed SLM and cloud-based LLM to power on-device agents with cost-effective reasoning.
- We propose a self-reinforced training method to boost the SLM’s task allocation accuracy without additional human annotation, and also preserve the general reasoning capabilities of SLM with a detachable, plug-and-play adapter.
- We conduct extensive experiments on a wide range of benchmarks to evaluate the effectiveness of our DoT framework and found that DoT can significantly reduce LLM costs while maintaining high reasoning accuracy.

## 2 Related Work

### 2.1 Reasoning with LLMs

Recent years have seen remarkable advancements in LLMs, with increasing model scale leading to powerful emergent reasoning capabilities. [37, 39]. Moreover, various prompt engineering techniques have been proposed to further extend LLMs’ reasoning capability. Chain-of-Thought (CoT) [38] improves reasoning performance by incorporating manually crafted decomposition steps into the prompt, allowing the LLM to follow the step-by-step resolution process. Building on this, Zero-shot CoT [18] achieves similar effects by simply adding the phrase "Let’s think step by step," enabling LLMs to automatically decompose and execute tasks. From the linear structure of CoT, more complex frameworks have been introduced, such as Tree-of-Thought [43] and Graph-of-Thought [2], which push the boundaries of reasoning through branching reasoning paths. However, these methods face increased resource demands and time complexity, and they heavily rely on manually predefined steps tailored to specific tasks, which significantly limits their generalizability. Additionally, there are some approaches, differing from the idea of predefined reasoning structures, that focus on how to decompose a complex problem into simpler ones for more effective resolution [17, 31, 32, 44].

Underlying these methods is the intuition of task decomposition and extensive empirical evidence has demonstrated its effectiveness. This inspires us to explore the potential of task decomposition in edge-cloud collaboration, allowing part of the task to be addressed by cloud-based models while the rest is handled by on-device models, thereby achieving a fine-grained collaborative framework.

### 2.2 On-device LLM Agent

The rise of LLMs has revolutionized AI applications, sparking interest in personal AI assistants and mobile automation tools. As on-device intelligent agents gain popularity, users expect seamless, real-time AI support on their smartphones [11]. However, the limited computational and storage capabilities of edge devices pose challenges in deploying powerful models for these agents. Under these constraints, edge-cloud collaboration is a practical solution. Apple’s latest research [12] exemplifies this synergy by combining an efficient on-device model, AFM-on-device, with a powerful cloud-based model, AFM-server. This approach balances device limitations with the high-performance needs of Apple’s AI features. Similarly, [6] addresses edge-device limitations by splitting task planning and execution between two models—Octo-planner and

与全尺寸大型语言模型（LLMs）相比，性能有所下降。初步研究也探索了边缘 - 云协作的潜力，其中超出本地部署的小型语言模型（SLMs）能力的任务会被重新路由到更强大的基于云的 LLMs [5, 13]。尽管如此，SLMs 与 LLMs 之间显著的性能差距通常导致推理能力与成本之间出现次优的权衡。

为了解决这个问题，我们的工作从经济学的基本概念“劳动分工”中获得灵感，该概念认为，将复杂任务分解为更小的组成部分通常能够带来更高效的解决方案，因为这可以让协作伙伴充分发挥各自的优势。在此基础上，我们提出了一种新颖的“思想分工”（**DoT**）框架，通过复杂的任务分解和优化的子任务分配，充分利用本地部署的 SLM 与云端 LLM 之间的协同效应。具体来说，DoT 利用一个任务分解器，这是一种元提示，将类似“思维链”的提示方法与精心策划的任务分解示例结合起来。这种方法能够发挥语言模型的内在规划能力，使它们能够将用户查询分解为更小的子任务。我们的核心见解是，即使是复杂的用户查询，通常也包含大量可以充分处理的简单子任务。

我们在七个广泛采用的 LLM 代理基准上进行了大量实验，涵盖多种场景，包括逻辑推理、网页浏览、数学问题求解和常识推理。结果表明，我们的 DoT 框架在显著降低 LLM 成本的同时，保持了与所有基准中最优基线方法相当的推理准确性。具体而言，与最准确的基线相比，平均推理时间和 API 成本分别降低了 66.12% 和 83.57%。此外，消融研究确认了我们关键模型设计选择的有效性。此外，与未利用“分解与分配”策略的任务转交方法相比，DoT 框架在成本 - 准确性权衡上始终表现优越，其中在低预算设置下性能增益尤其显著。我们进一步展示了 DoT 在不同基准上的强泛化能力。

总而言之，我们的贡献有三方面：

- 我们提出了一种新颖的思维分工（DoT）框架，充分利用本地部署的 SLM 与基于云的 LLM 之间的协同作用，为设备端智能体提供具有成本效益的推理能力。
- 我们提出了一种自我强化的训练方法，以在无需额外人工标注的情况下提升 SLM 的任务分配准确性，同时通过可拆卸、即插即用的适配器保留 SLM 的通用推理能力。
- 我们在各种基准上进行了广泛的实验，以评估我们 DoT 框架的有效性，并发现 DoT 在保持高推理准确性的同时可以显著降低大型语言模型的成本。

## 2 相关工作

### 2.1 使用大型语言模型进行推理

近年来，大型语言模型（LLMs）取得了显著进展，模型规模的不断扩大带来了强大的新兴推理能力。[37, 39]。此外，各种提示工程技术被提出以进一步扩展 LLMs 的推理能力。链式思维（Chain-of-Thought, CoT）[38] 通过在提示中加入手工设计的分解步骤来提高推理性能，使 LLM 能够遵循逐步解决问题的过程。在此基础上，零样本链式思维（Zero-shot CoT）[18] 仅需添加“让我们一步一步思考”这句话便能实现类似效果，使 LLM 能够自动分解和执行任务。基于 CoT 的线性结构，引入了更复杂的框架，如思维树（Tree-of-Thought）[43] 和思维图（Graph-of-Thought）[2]，通过分支推理路径推动推理能力边界的扩展。然而，这些方法面临更高的资源需求 and 时间复杂度，并且高度依赖手工预定义的针对性步骤。

这些方法的基础是任务分解的直觉，大量实证研究已经证明了其有效性。这激励我们探索任务分解在边缘 - 云协作中的潜力，允许部分任务由基于云的模型处理，而其余部分由设备端模型处理，从而实现一个细粒度的协作框架。

### 2.2 设备端 LLM 代理

大型语言模型（LLM）的兴起彻底改变了人工智能的应用，引发了个人 AI 助手和移动自动化工具的关注。随着设备端智能代理的普及，用户期望在智能手机上获得无缝的实时 AI 支持 [11]。然而，边缘设备有限的计算和存储能力在为这些代理部署强大模型时带来了挑战。在这些限制下，边缘 - 云协作是一个切实可行的解决方案。苹果公司最新的研究 [12] 通过将高效的设备端模型 AFM-on-device 与强大的云端模型 AFM-server 结合，体现了这种协同。这种方法平衡了设备的限制与苹果 AI 功能对高性能的需求。类似地，[6] 通过将任务规划和执行分配给两个模型——Octo-planner 和

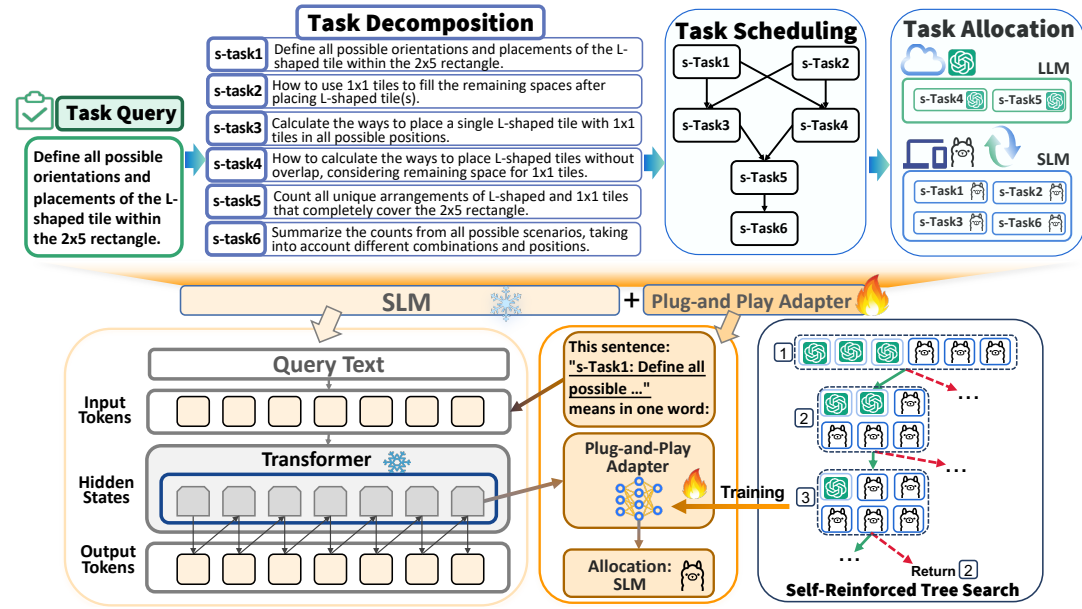


Figure 1: Overview of Our Proposed DoT Framework.

Octopus—focused on efficiency and adaptability. Both approaches highlight the trend of edge-cloud collaboration to ensure powerful, low-latency AI experiences, though Apple leans more on cloud support while Octo-planner emphasizes on-device optimization.

### 3 Preliminaries

**Problem Definition** Denote the local deployed SLM as  $M_D$ , and the cloud-based LLM as  $M_C$ . The user's original query is restricted to the edge model for task decomposition and allocation, while the resulting sub-tasks can be resolved by either  $M_D$  or  $M_C$ . The entire set of reasoning tasks is represented as  $\mathcal{T} = \{T_1, T_2, \dots, T_n\}$ . Let the reasoning accuracy over the entire task set be denoted as  $Acc$ , with the API cost represented by  $C_{Api}$ , and the completion time denoted as  $C_{Time}$ . For each task  $T$ , denote the decomposition process as:

$$T \rightarrow \{t^1, t^2, \dots, t^k\} \quad (1)$$

Based on the decomposed subtasks  $t^i$ , the model allocation scheme can be denoted as:

$$M : t^i \mapsto \{M_D, M_C\} \quad (2)$$

which prioritizes assigning simple subtasks to on-device SLM, while invoking the cloud-based LLM for handling complex subtasks.

The goal of our optimization is to minimize the discrepancy between the model's allocation scheme  $M$  and the optimal scheme  $M^*$ :

$$\min |M - M^*| \quad (3)$$

The optimal scheme  $M^*$  is derived through a search strategy that maximizes SLM usage while maintaining accuracy. During the optimization process, as the allocation scheme gradually approaches the optimal solution, both time cost  $C_{Time}$  and API cost  $C_{Api}$  decrease, while  $Acc$  remains well-maintained.

## 4 Methodology

### 4.1 Division-of-Thoughts Framework

The *division-of-labour* is a prominent concept in economics, famously introduced by Adam Smith. It refers to the separation and allocation of tasks within any economic system or organization, enabling participants to specialize in specific areas based on their unique capabilities. This specialization allows individuals, organizations, and nations to optimize productivity by leveraging specialized skills, equipment, and resources. This concept inspires us to consider a similar approach in the context of edge-cloud collaboration, where user queries can be intricately divided: simpler sub-tasks can be assigned to SLMs while more complex ones are allocated to LLMs. This sub-task-level division of labor is expected to reduce reasoning costs while maintaining reasoning performance, enabling more efficient collaboration.

Building on this intuition, we develop the **DoT** edge-cloud collaboration framework. As illustrated in Figure 1, our framework is divided into three components. The first component is the *Task Decomposer*, which breaks down the user's query into several simpler and independent sub-tasks. The second component is the *Task Scheduler*, which is leveraged to determine the pairwise dependencies between sub-tasks and constructing a task dependency graph based on these dependencies. The third component is the *plug-and-play LLM adapter*, responsible for assigning each sub-task to the appropriate models. The adapter extracts sentence embeddings from the SLM and maps them to difficulty coefficients, which serve as the basis for the model allocation for sub-tasks. Importantly, the training of the adapter does not require modifying the LLM's parameters, ensuring that the LLM's question-answering remains unaffected. Once the appropriate models have been assigned to each sub-task, reasoning proceeds along the order defined by the constructed dependency graph, leading to the final results. Figure 2

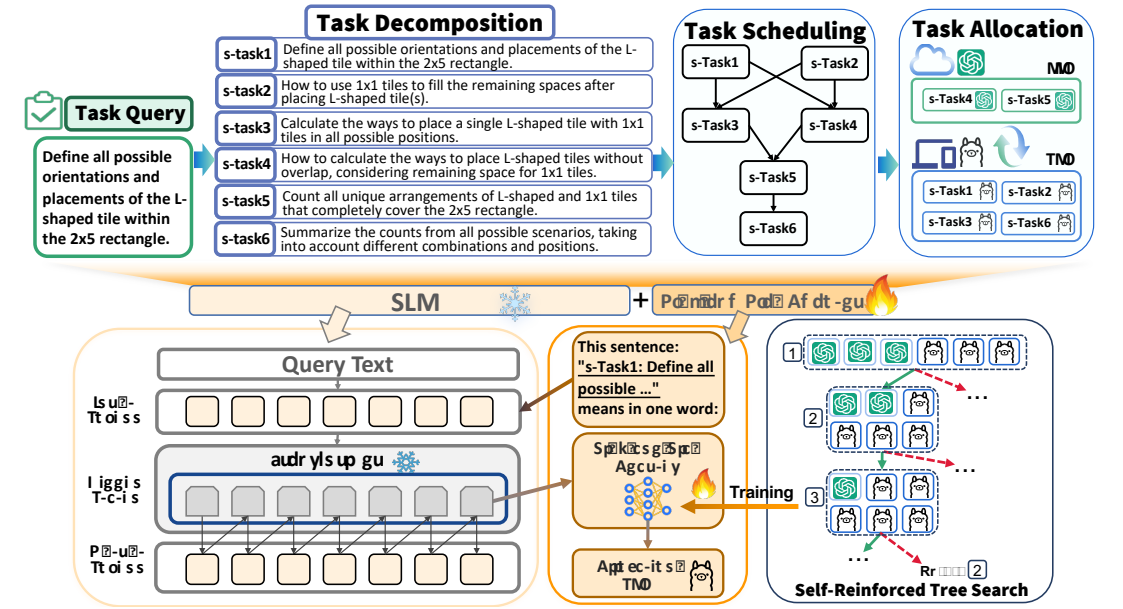


图 1: 我们提出的 DoT 框架概览。

章鱼——专注于效率和适应性。这两种方法都突出了边缘与云协作的趋势，以确保强大、低延迟的人工智能体验，尽管苹果更多依赖云支持，而章鱼规划者则强调设备端优化。

### 3 预备事项

**问题定义** 将本地部署的 SLM 记作  $M_D$ ，云端的 LLM 记作  $M_C$ 。用户的原始查询限于边缘模型进行任务分解和分配，而生成的子任务可以由  $M_D$  或  $M_C$  解决。整个推理任务集合表示为  $\mathcal{T} = \{T_1, T_2, \dots, T_n\}$ 。设整个任务集合的推理准确性记作  $Acc$ ，API 成本记作  $C_{Api}$ ，完成时间记作  $C_{Time}$ 。对于每个任务  $T$ ，将分解过程表示为：

$$T \rightarrow \{t^1, t^2, \dots, t^k\} \quad (1)$$

基于分解后的子任务  $t^i$ ，模型分配方案可以表示为：

$$M : t^i \mapsto \{M_D, M_C\} \quad (2)$$

它优先将简单子任务分配给设备上的 SLM，同时调用基于云的 LLM 来处理复杂子任务。

我们优化的目标是最小化模型的分配方案  $M$  与最优方案  $M^*$  之间的差异：

$$\min |M - M^*| \quad (3)$$

通过一种搜索策略导出最优方案，该策略在保持准确性的同时最大化 SLM 的使用。在优化过程中，随着分配方案逐渐接近最优解，时间成本和 API 成本下降，而仍然保持良好。

## 4 方法论

### 4.1 思维划分框架

劳动力分工是经济学中的一个重要概念，由亚当·斯密著名地提出。它指的是在任何经济系统或组织中任务的分离和分配，使参与者能够根据自身的独特能力在特定领域进行专业化。这种专业化使个人、组织和国家能够通过利用专业技能、设备和资源来优化生产力。这个概念启发我们在边缘-云协作的背景下考虑类似的方法，其中用户查询可以被精细地分解：较简单的子任务可以分配给 SLM，而更复杂的任务则分配给 LLM。这种子任务级的劳动力分工预计能够在保持推理性能的同时降低推理成本，从而实现更高效的协作。

在此直觉的基础上，我们开发了 **DoT** 边缘-云协作框架。如图 1 所示，我们的框架分为三个部分。第一部分是任务分解器，它将用户的查询拆解为几个更简单且相互独立的子任务。第二部分是任务调度器，用于确定子任务之间的成对依赖关系，并基于这些依赖关系构建任务依赖图。第三部分是即插即用 LLM 适配器，负责将每个子任务分配给合适的模型。适配器从 SLM 中提取句子嵌入并将其映射到难度系数，这些系数作为子任务模型分配的基础。重要的是，适配器的训练不需要修改 LLM 的参数，从而确保 LLM 的问答功能不受影响。一旦为每个子任务分配了合适的模型

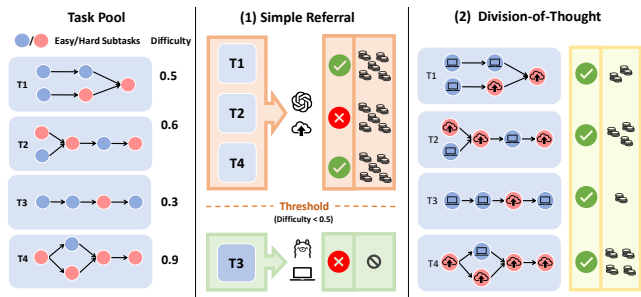


Figure 2: Advantage of “Division-and-Allocate” Strategy

clearly demonstrates that, compared to the simple referral strategy, DoT exhibits significant advantages in both accuracy and efficiency.

## 4.2 Decomposing User Query

As emphasized in the *division-of-labour*, an effective division method serves as the foundation for collaborative work. The granularity and accuracy of division directly influence the quality and efficiency of the collaboration. Prior research on model collaboration has primarily focused on query-level granularity and relied heavily on manual task decomposition strategies. A typical example is the ToT approach in Game-of-24, which employs a fixed two-step process of proposal generation and evaluation. This rigid, manually-designed workflow lacks generalizability across diverse tasks, underscoring the necessity for automated, flexible, and fine-grained task decomposition methods.

To address these challenges, we develop a fine-grained task decomposition method based on the powerful *In-Context Learning (ICL)* capabilities of LLMs. We leverage a meta-prompt that incorporates “chain-of-thought”-like prompting with hand-crafted task decomposing examples, to elicit the inherent *planning* abilities in SLMs. For each benchmark, We randomly selected 8 samples, manually performed step-by-step task decomposition, and incorporated these steps into the prompts. To enable LLMs to independently solve each sub-task and effectively use the answers from preceding tasks as references for subsequent reasoning, we place great emphasis on the independence and clarity of the decomposed sub-tasks and have incorporated targeted cues in the prompts design.

## 4.3 Task Scheduling via Dependency Graph

We employ a three-step approach to schedule all the sub-tasks effectively: **Dependency Judgement**. Prompting LLM agent to assess pairwise dependencies between sub-tasks. The prompts are as follows: *"Please list the dependencies in the format 'Subproblem A [xxx] -> Subproblem B [xxx]' indicating that Subproblem A must be completed before Subproblem B can start."* **Graph Construction**. Converting dependencies into a dependency graph. Based on the clear directional dependencies, constructing the graph is straightforward. However, we further trace back from the final sub-task node to calculate the depth of each sub-task node in the graph (analogous to the height in a tree structure). Sub-tasks at the same depth are independent of each other and can be inferred in parallel. Depth serves as the inference batch, with batches processed sequentially while tasks within a batch are reasoned in parallel. As shown in Figure 3, compared to sequentially reasoning through

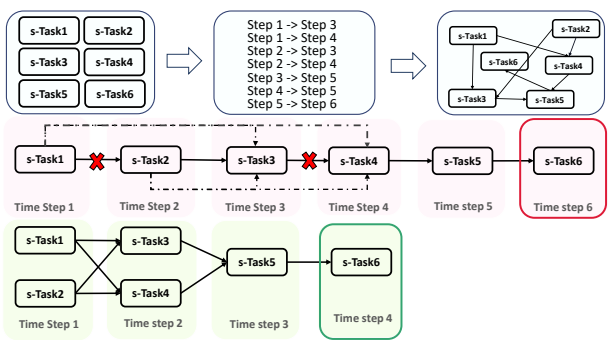


Figure 3: Illustrating Dependency Graph of Task Scheduling

all sub-tasks, our graph structure can capture more precise inferential relationships while incurring fewer time costs. **On-Graph Reasoning**. We start by solving the sub-tasks from the shallowest depth batch, running the sub-tasks within the same batch in parallel. After completing one batch, we proceed to the next. During the reasoning of a specific sub-task, only the results of its prerequisite sub-tasks, rather than all previously solved tasks, are included in the prompts, which enables efficient graph-based reasoning.

## 4.4 Task Allocation with Reinforced SLM

For the decomposed sub-tasks, we aim to assess their difficulty based on the task descriptions and allocate either cloud-based or edge-side models for execution. Using LLMs to evaluate difficulty introduces additional inference costs and often fails to accurately assess the sub-task’s complexity, which motivates us to train a model specifically designed for task allocation with sentence embeddings. *Sentence embeddings*, which map a sentence to a fixed-size vector capturing its semantic meaning and context, have seen extensive application in natural language processing for their lightweight accessibility and the strong ability to capture sentence semantics. We can obtain a sentence embedding for each sub-task’s prompt and then map the embedding to the corresponding difficulty. Given that LLMs are already deployed on the edge side and have been shown to serve as effective sentence embedders, we plan to leverage the local deployed SLM to produce sentence embeddings.

However, autoregressive LLMs lack specialized tokens such as BERT’s [MASK] or [CLS], which are typically used in transformer-based models for embedding tasks. The exploration of Ting Jiang et al. [15] helps us overcome this limitation. Prompt-based method is refined specifically for autoregressive LLMs by instructing the model to generate the next word that captures the semantic meaning of the input sentence. Specifically, a simple but effective template can be constructed for each piece of text: *This sentence: "[text]" means in one word: "*, where *[text]* represents the input sentence, and the LLM is prompted to generate the next token that encapsulates the core meaning of the sentence in a single word. The last generated token plays a critical role, as we extract its hidden state from the model and use it as the sentence embedding.

**Boosting SLM with Plug-and-Play Adapter**. Despite the varying lengths of sub-task descriptions, the sentence embeddings maintain a consistent dimensionality. For efficiency, we append a multi-layer perceptron (MLP) to the transformer module of the LLM to map the embeddings to a difficulty score. Given that both the edge

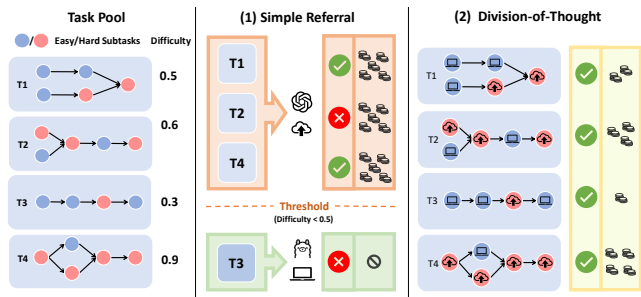


图 2: “分而配”策略的优势

清楚地表明, 与简单的推荐策略相比, DoT 在准确性和效率方面都表现出显著优势。

## 4.2 分解用户查询

正如在劳动分工中所强调的, 一个有效的分工方法是协作工作的基础。分工的粒度和准确性直接影响协作的质量和效率。以往关于模型协作的研究主要集中在查询级粒度, 并严重依赖人工任务分解策略。一个典型的例子是在 24 点游戏中的 ToT 方法, 它采用固定的两步流程: 提议生成和评估。这种僵硬的、人工设计的工作流程在不同任务之间缺乏通用性, 凸显了自动化、灵活和精细化任务分解方法的必要性。

为了解决这些挑战, 我们开发了一种基于大型语言模型 (LLMs) 强大上下文学习 (ICL) 能力的细粒度任务分解方法。我们利用一个元提示 (meta-prompt), 其中结合了类似 “思路链” (chain-of-thought) 的提示与手工制作的任务分解示例, 以引出 SLMs 内在的规划能力。对于每个基准测试, 我们随机选择了 8 个样本, 手动执行逐步任务分解, 并将这些步骤纳入提示中。为了使 LLMs 能够独立解决每个子任务, 并有效利用前面任务的答案作为后续推理的参考, 我们非常重视分解子任务的独立性和清晰性, 并在提示设计中加入了针对性的提示。

## 4.3 通过依赖图进行任务调度

我们采用三步法来有效地安排所有子任务: **依赖判断**。提示大语言模型 (LLM) 代理评估子任务之间的成对依赖关系。提示如下: “请以 ‘子问题 A [xxx] -> 子问题 B [xxx]’ 的格式列出依赖关系, 表示子问题 A 必须在子问题 B 开始之前完成。” **图构建**。将依赖关系转换为依赖图。根据明确的定向依赖关系, 构建图是直接的。然而, 我们进一步从最终子任务节点回溯, 以计算图中每个子任务节点的深度 (类似于树结构中的高度)。处于同一深度的子任务相互独立, 可以并行推理。深度作为推理批次, 每个批次按顺序处理, 而批次内的任务

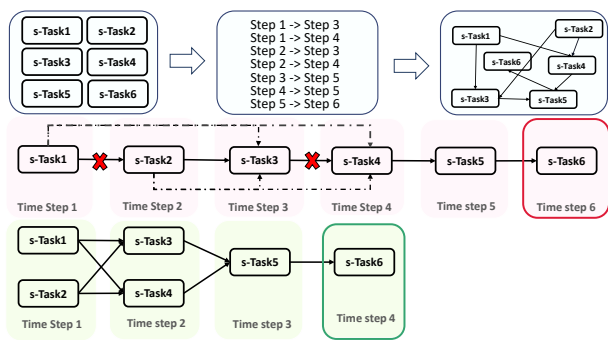


图 3: 任务调度的依赖关系图示

对于所有子任务, 我们的图结构可以在降低时间成本的同时捕捉更精确的推理关系。**图上推理**。我们从解决最浅深度批次的子任务开始, 并在同一批次内并行运行子任务。完成一个批次后, 我们继续下一个批次。在推理特定子任务时, 仅将其前置子任务的结果包含在提示中, 而不是所有已解决的任务, 这使得基于图的高效推理成为可能。

## 4.4 使用增强 SLM 的任务分配

对于分解后的子任务, 我们旨在根据任务描述评估其难度, 并为执行分配基于云端或边缘端的模型。使用大型语言模型 (LLM) 来评估难度会带来额外的推理成本, 并且常常无法准确评估子任务的复杂性, 这促使我们专门训练一个用于任务分配的模型, 该模型使用句子嵌入。句子嵌入将句子映射到固定大小的向量, 以捕捉其语义含义和上下文, 在自然语言处理中的应用非常广泛, 因为它们轻量且能够强有力地捕捉句子语义。我们可以为每个子任务的提示获取句子嵌入, 然后将该嵌入映射到相应的难度。鉴于大型语言模型已经部署在边缘端, 并且已被证明可作为有效的句子嵌入器, 我们计划利用本地部署的 SLM 生成句子嵌入。

然而, 自回归大语言模型 (LLM) 缺乏像 BERT 的 [MASK] 或 [CLS], 这样的专用标记, 而这些标记通常用于基于 Transformer 的模型中的嵌入任务。Ting Jiang 等人的研究 [15] 帮助我们克服了这一限制。基于提示的方法专门为自回归 LLM 优化, 通过指导模型生成捕捉输入句子语义的下一个词汇。具体来说, 可以为每段文本构建一个简单但有效的模板: 这个句子: "[text]" 的意思用一个词来说是: ", 其中 [text] 表示输入句子, LLM 被提示生成下一个标记, 该标记用一个词概括句子的核心意思。最后生成的标记起着关键作用, 因为我们会从模型中提取其隐藏状态并将其用作句子嵌入。

**通过即插即用适配器提升 SLM**。尽管子任务描述的长度各不相同, 句子嵌入仍保持一致的维度。为了提高效率, 我们在大型语言模型 (LLM) 的变压器模块上附加了一个多层感知器 (MLP), 用于将嵌入映射到难度分数。鉴于两者的边界

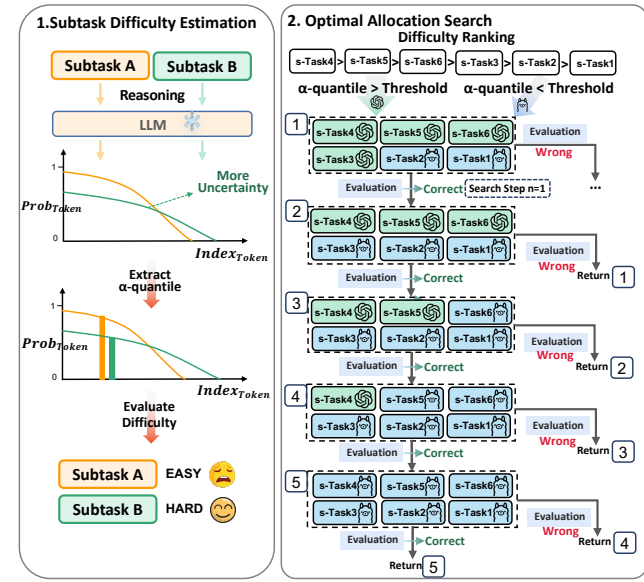


Figure 4: Tree Search-Based Dataset Construction Process

side and the cloud host only one model, we can simply set the scores as 0 (simple, for SLM) or 1 (complex, for LLM). The output from the MLP is translated into a model designation and passed to the agent handling the current reasoning step, providing a concrete allocation strategy for edge-cloud collaboration. Moreover, the adapter requires no modifications to SLM's parameters. It essentially serves as a flexible and extensible adapter for the SLM, avoiding the need to store two sets of parameters on the device.

**Creating Training Data with  $\alpha$ -Tree Algorithm.** How can we construct a dataset to train our adapter? We develop an efficient allocation optimization strategy,  $\alpha$ -Tree, which consists of two key components: the sub-task difficulty ranking based on the  $\alpha$ -quantile token probability, and a tree-based search strategy guided by the ranking, allowing us to generate a large-scale optimal allocation dataset with low cost and high speed.  $\alpha$ -quantile refers to calculating a specific quantile from the token probabilities generated by LLM during inference. Unlike traditional methods that aggregate probabilities through summation or averaging,  $\alpha$ -quantile focuses on a specific portion of the uncertainty distribution, such as the minimum ( $\alpha = 0$ ) or a higher percentile (e.g.,  $\alpha = 0.8$ ), which provides a fine-grained measure of uncertainty. Denote the input context as  $x$ , and the output tokens as  $\hat{y}_i$ . The  $\alpha$ -quantile value is denoted as:

$$s_{\text{quant}}(x, \alpha) \doteq \text{quantile}_{\alpha} \left( p^{(1)}(\hat{y}'_1 | x), p(\hat{y}'_2 | x, \hat{y}'_1), \dots, p(\hat{y}'_n | x, \hat{y}'_1, \dots, \hat{y}'_{n-1}) \right) \quad (4)$$

In our implementation, we employ the SLM to answer all decomposed sub-questions, recording the sampling probability corresponding to each token. We then apply a consistent alpha value to extract the  $\alpha$ -quantile from each probability sequence and ranked the  $\alpha$ -quantile values for all sub-task responses. Since higher task difficulty leads to greater uncertainty in the model's answers, resulting in lower token sampling probabilities, we reverse the  $\alpha$ -quantile ranking to obtain the difficulty order of the sub-tasks.

### Algorithm 1 Self-reinforced dataset construction method: $\alpha$ -Tree

**Input:** Tasks which has been decomposed into several subtask:

$$\{T_1 = \{t_1^1, \dots, t_1^{k_1}\}, T_2 = \{t_2^1, \dots, t_2^{k_2}\}, \dots, T_n = \{t_n^1, \dots, t_n^{k_n}\}\}$$

**Output:** Fine-grained device-cloud model allocation scheme applied to each subtask within every task ( $m_i^j \in \{M_{\mathcal{D}}, M_{\mathcal{C}}\}$ ):

$$\{M_1 = \{m_1^1, \dots, m_1^{k_1}\}, M_2 = \{m_2^1, \dots, m_2^{k_2}\}, \dots, M_n = \{m_n^1, \dots, m_n^{k_n}\}\}$$

- 1: **for** task id  $i$  **do**
- 2:   Perform on-graph reasoning for each subtask:  $\text{Answer} = \{a_i^1, \dots, a_i^{k_i}\}$  with  $\text{Prob}_{\text{token}} = \{p_i^1, \dots, p_i^{k_i}\}$
- 3:   Calculate the  $\alpha$ -quantile value for each token sequence:  $V_{\alpha\text{-quantile}} = \{v_i^1, v_i^2, \dots, v_i^{k_i}\}$
- 4:   Set a threshold  $\theta$  to obtain the initial allocation strategy:  $M_i = \{m_i^1, \dots, m_i^{k_i}\}$  where  $m_i^j = \begin{cases} M_{\mathcal{D}} & \text{if } v_i^j > \theta \\ M_{\mathcal{C}} & \text{if } v_i^j < \theta \end{cases}$
- 5:   Reasoning on the task to obtain:  $\text{result} = \text{True or False}$
- 6:   **while** True **do**
- 7:     **if**  $\text{result} == \text{True}$  **then**
- 8:       Find all subtasks assigned to  $M_{\mathcal{C}}$ ,  $M_{\mathcal{C}i} = \{m \in M_i \mid m == M_{\mathcal{C}}\}$ , select N subtasks with the highest  $\alpha$ -quantile probabilities, and reassign them to  $\mathcal{D}$ . Get new  $M'_i$ .
- 9:     **else**
- 10:       Find all subtasks assigned to  $M_{\mathcal{D}}$ ,  $M_{\mathcal{D}i} = \{m \in M_i \mid m == M_{\mathcal{D}}\}$ , select N subtasks with the lowest  $\alpha$ -quantile probabilities, and reassign them to  $M_{\mathcal{C}}$ . Get new  $M'_i$ .
- 11:     **end if**
- 12:     Reasoning to obtain new result  $\text{result}'$
- 13:     **if**  $\text{len}(M_{\mathcal{D}i}) == 0$  or  $\text{len}(M_{\mathcal{C}i}) == 0$  or  $\text{result}' \neq \text{result}$  **then**
- 14:       break
- 15:     **end if**
- 16:      $M_i = M'_i$
- 17:   **end while**
- 18: **end for**

Based on the  $\alpha$ -quantile values of the sub-tasks, we set a fixed allocation threshold: sub-tasks with values exceeding this threshold are assigned to SLM, while those below the threshold are assigned to LLM, forming the initial model allocation. From this initial allocation, we perform collaborative reasoning with SLM and LLM. If the reasoning result is correct, we reassign  $n$  of the sub-tasks currently handled by LLM, specifically those with the highest probability of being correctly answered, to SLM. Conversely, if the answer is incorrect, we transfer  $n$  of the sub-tasks handled by SLM, specifically those with the lowest probability of being misanswered, to LLM. As shown in Figure 4, the structure of the search exhibits a tree-like form. The algorithmic workflow for dataset construction is illustrated in Algorithm 1.

## 5 Experiments

### 5.1 Experimental Setup

**Benchmarks.** We validate the effectiveness of our framework across seven open-source benchmarks. For each benchmark, we randomly select 8 instances for manual annotation to serve as in-context learning examples of task decomposition. Additionally, we

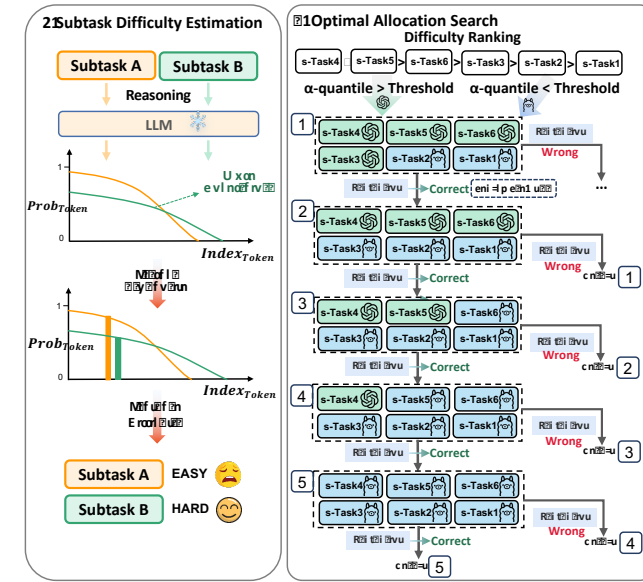


图 4：基于树搜索的数据集构建过程

在边缘侧和云端只运行一个模型的情况下，我们可以简单地将分数设置为 0（简单，适用于 SLM）或 1（复杂，适用于 LLM）。MLP 的输出会被转换为模型标识，并传递给处理当前推理步骤的代理，从而提供边缘 - 云协作的具体分配策略。此外，适配器不需要修改 SLM 的参数。它本质上作为 SLM 的一个灵活且可扩展的适配器，避免了在设备上存储两套参数的需求。

**使用  $\alpha$ -树算法创建训练数据。**我们如何构建数据集来训练我们的适配器？我们开发了一种高效的分配优化策略， $\alpha$ -Tree，它包括两个关键组件：基于  $\alpha$ -分位数令牌概率的子任务难度排名，以及由排名引导的基于树的搜索策略，使我们能够以低成本和高速生成大规模的最优分配数据集。 $\alpha$ -分位数指的是计算 LLM 在推理过程中生成的令牌概率的特定分位数。与通过求和或平均聚合概率的传统方法不同， $\alpha$ -分位数关注不确定性分布的特定部分，例如最小值（ $\alpha = 0$ ）或较高的百分位数（如  $\alpha = 0.8$ ），这提供了不确定性的精细测量。将输入上下文记为  $x$ ，输出令牌记为  $\hat{y}_i$ 。 $\alpha$ -分位数值表示

$$s_{\text{quant}}(x, \alpha) \doteq \text{quantile}_{\alpha} \left( p^{(1)}(\hat{y}'_1 | x), p(\hat{y}'_2 | x, \hat{y}'_1), \dots, p(\hat{y}'_n | x, \hat{y}'_1, \dots, \hat{y}'_{n-1}) \right) \quad (4)$$

在我们的实现中，我们使用 SLM 来回答所有分解后的子问题，并记录每个标记对应的采样概率。然后我们应用一致的 alpha 值从每个概率序列中提取  $\alpha$  分位数，并对所有子任务回答的  $\alpha$  分位数值进行排序。由于任务难度较高会导致模型答案的不确定性增加，从而导致标记采样概率降低，我们反转  $\alpha$  分位数排名以获得子任务的难度顺序。

**算法 1 自增强数据集构建方法：** $\alpha$ -Tree 输入：已分解为多个子任务的任务： $\{T_1 = \{t_1^1, \dots, t_1^{k_1}\}, T_2 = \{t_2^1, \dots, t_2^{k_2}\}, \dots, T_n = \{t_n^1, \dots, t_n^{k_n}\}\}$  输出：应用于每个任务中每个子任务的细粒度设备 - 云模型分配方案 ( $m_i^j \in \{M_{\mathcal{D}}, M_{\mathcal{C}}\}$ ):  $\{M_1 = \{m_1^1, \dots, m_1^{k_1}\}, M_2 = \{m_2^1, \dots, m_2^{k_2}\}, \dots, M_n = \{m_n^1, \dots, m_n^{k_n}\}\}$  1: 对于任务 id  $i$  执行 2: 对每个子任务执行图上推理:  $\text{Answer} = \{a_i^1, \dots, a_i^{k_i}\}$  使用  $\text{Prob}_{\text{token}} = \{p_i^1, \dots, p_i^{k_i}\}$  计算每个标记序列的  $\alpha$  分位值:  $V_{\alpha\text{-quantile}} = \{v_i^1, v_i^2, \dots, v_i^{k_i}\}$  3: 设置阈值  $\theta$  以获得初始分配策略:  $M_i = \{m_i^1, \dots, m_i^{k_i}\}$  其中  $m_i^j = \begin{cases} M_{\mathcal{D}} & \text{如果 } v_i^j > \theta \\ M_{\mathcal{C}} & \text{如果 } v_i^j < \theta \end{cases}$  4: 对任务进行推理以获得: 结果 = 真或假 5: 当 True 时执行 6: 如果结果 == 为真, 则 7: 找到分配给  $M_{\mathcal{C}}$ ,  $M_{\mathcal{C}i} = \{m \in M_i \mid m == M_{\mathcal{C}}\}$  的所有子任务, 选择具有最高  $\alpha$  分位概率的 N 个子任务, 并重新分配给  $\mathcal{D}$  8: 找到分配给  $M_{\mathcal{D}}$ ,  $M_{\mathcal{D}i} = \{m \in M_i \mid m == M_{\mathcal{D}}\}$  的所有子任务, 选择具有最低  $\alpha$  分位概率的 N 个子任务, 并重新分配给  $M_{\mathcal{C}}$  9: 结束 10: 结束

| Model            | Logical Reasoning |         |         |            |         |         | Web Browsing |         |         | Solving Math Problems |         |         |            |         |         |            |         |         | Commonsense Reasoning |         |         |
|------------------|-------------------|---------|---------|------------|---------|---------|--------------|---------|---------|-----------------------|---------|---------|------------|---------|---------|------------|---------|---------|-----------------------|---------|---------|
|                  | P3                |         |         | SCAN       |         |         | WebShop      |         |         | MATH                  |         |         | CHAMP      |         |         | DROP       |         |         | CSQA                  |         |         |
|                  | Acc               | CTime   | C_API   | Acc        | CTime   | C_API   | Acc          | CTime   | C_API   | Acc                   | CTime   | C_API   | Acc        | CTime   | C_API   | Acc        | CTime   | C_API   | Acc                   | CTime   | C_API   |
| COT (GPT-4o)     | <b>42%</b>        | 35.8    | 4.45€   | <b>68%</b> | 9.21    | 2.75€   | 35%          | 30.9    | 10.65€  | 51.5%                 | 34.5    | 5.34€   | 55.5%      | 26.4    | 4.45€   | 80%        | 11.6    | 1.30€   | 80%                   | 17.0    | 3.60€   |
| TOT (GPT-4o)     | 38%               | 93.1    | 14.55€  | 52%        | 32.5    | 9.82€   | <b>36%</b>   | 62.4    | 47.34€  | <b>63%</b>            | 60.5    | 9.97€   | 57%        | 64.2    | 11.65€  | 80.5%      | 40.2    | 5.41€   | 82%                   | 98.8    | 20.50€  |
| COT (Llama 3-8B) | 5.5%              | 18.1    | N/A     | 17%        | 5.0     | N/A     | 0.0%         | 10.5    | N/A     | 10%                   | 21.1    | N/A     | 19%        | 13.1    | N/A     | 72%        | 3.8     | N/A     | 70%                   | 8.4     | N/A     |
| TOT (Llama 3-8B) | 5.5%              | 58.3    | N/A     | 13%        | 21.8    | N/A     | 1.4%         | 22.5    | N/A     | 29.5%                 | 49.0    | N/A     | 25%        | 68.1    | N/A     | 65%        | 27.8    | N/A     | 68.5%                 | 89.4    | N/A     |
| DataShunt        | 14%               | 25.1    | 2.45€   | 23.5%      | 7.6     | 1.72€   | 34%          | 30.9    | 8.35€   | 16%                   | 24.9    | 1.66€   | 34%        | 19.1    | 2.98€   | 74%        | 8.6     | 0.60€   | 73%                   | 10.4    | 1.28€   |
| DoT (ours)       | 41%               | 23.5    | 1.58€   | 63%        | 5.5     | 1.20€   | 31%          | 17.2    | 4.97€   | 59%                   | 22.6    | 1.02€   | <b>58%</b> | 16.1    | 0.84€   | <b>85%</b> | 4.9     | 0.32€   | <b>82%</b>            | 9.9     | 0.49€   |
| Improvement      | ↓2.38%            | ↓34.36% | ↓64.49% | ↓7.35%     | ↓40.28% | ↓56.36% | ↓13.89%      | ↓72.43% | ↓89.95% | ↓6.35%                | ↓62.72% | ↓89.50% | ↑1.75%     | ↓74.92% | ↓92.79% | ↑5.59%     | ↓87.81% | ↓94.09% | 0%                    | ↓89.98% | ↓97.60% |

Table 1: Performance of DoT and baselines on 7 benchmarks. CTime and C\_API are averaged expense for each task, where time consumption is measured in seconds, and API cost is measured in US dollar cents (¢). N/A appears in experiments where reasoning is conducted solely using LLaMA without invoking the OpenAI’s API key. In each benchmark, the highest reasoning accuracy is highlighted in bold. The results of the baseline with the highest Acc are underlined which will be used to compute the "Improvement" in the last row.

randomly extract 200 questions from each benchmark to constitute our test set. The benchmarks are categorized into four groups:

- **Logical Reasoning.** We select P3 [30] and SCAN [20] for evaluation. These benchmarks place a stronger emphasis on logic, involving challenges such as traversal, backward reasoning, and anomaly detection, which require a higher level of logical coherence and rigor.
- **Math Problems Solving.** This category of benchmarks primarily addresses mathematical problems, involving computation, mathematical knowledge, and problem-solving techniques. It is widely regarded as a crucial test of LLMs’ reasoning abilities. We select three benchmarks for this category: MATH [14], CHAMP [25], and DROP [9].
- **Commonsense Reasoning.** CSQA [36] is a widely used commonsense reasoning dataset that places less emphasis on reasoning capabilities but requires a broader knowledge base. The difference in LLMs’ parameter scales leads to disparities in the knowledge systems they possess, making CSQA well-suited for evaluating our collaborative reasoning scenario.
- **Web Browsing.** We choose WebShop [42] which tests LLMs as interactive agents in real-world scenarios. It challenges LLM agents to navigate web pages, interpret complex queries, and take appropriate actions to fulfill user requests. This benchmark closely aligns with AI assistant scenarios, encompassing instructions understanding, query reformulation, and strategic information exploration.

**Baselines and Prompting Methods.** Given the constraints of the edge-cloud collaborative setting, we establish three baselines.

- **CoT [38].** This baseline decomposes the problem into sub-steps, solving each sequentially. We use a single LLM throughout, executing each step only once without iteration.
- **ToT [43].** ToT explores multiple solutions (M=5) for each sub-step. We use a scoring mechanism to retain the N=3 top-scored paths, with the highest-scored path determining the final answer. ToT also employs a single LLM for reasoning.
- **DataShunt [3].** This approach dynamically selects between on-device SLM and cloud-based LLM at the start of each task. It prompts the LLM to evaluate task complexity, assigning a *solving model* that is then used consistently for all sub-steps.

In the task decomposition processes involved in CoT and ToT, we employed the same prompting method used in our DoT, applying

eight hand-crafted examples to each benchmark. These examples are randomly sampled from the dataset and are orthogonal to the test set. The specific prompts can be found in Appendix B.5.

**Selection and Deployment of LLMs** We used GPT-4o [29] as cloud-based LLM and Llama 3-8B [27] as on-device LLM as its parameter scale is theoretically deployable on edge devices. The Llama 3-8B model is deployed on a local A100 GPU with a context length limit of 8192. The parameter count of our adapter is only 13,109,249, approximately 1/800th of the parameters in local Llama.

**Evaluation.** For evaluation, we establish three metrics: *Acc*, *Ctime*, *C\_API*. Six of the benchmarks, excluding WebShop, have deterministic results. For instance, the results for CSQA are single-letter options, allowing direct determination of correctness. Similarly, for P3, the reasoning output can be passed into a problem function, and if the function returns True, the reasoning is deemed correct. For WebShop, we use the evaluation environment provided by the original benchmark to average the reward of all purchased items on the level of satisfaction according to the constraints.

## 5.2 Main Results

Comparison between our method and baselines is shown in Table 1, we have highlighted in bold the highest accuracy results among the five baseline experiments on each benchmark, while the associated costs are underlined. We compute the relative improvement of our results compared to the baseline with the highest accuracy.

The experimental results demonstrate that our approach significantly reduces costs while maintaining accuracy within an acceptable range of decline. Across seven benchmarks, the relative changes in accuracy compared to the best baseline results are: -2.38%, -7.35%, -13.89%, -6.35%, 1.75%, 5.59%, and 0%. Notably, positive improvements are observed in some cases, suggesting that our framework has the potential to enhance LLMs’ reasoning capability. This enhancement is further discussed in ablation study 5.4. At the same time, our approach achieves a substantial reduction in cost compared to the baseline with the highest accuracy, with an average time reduction of 66.12% and an average API cost reduction of 83.57% which far exceeded the decline in accuracy.

Moreover, on P3 and SCAN benchmarks, CoT achieves the best performance, indicating that a sequential linear reasoning structure is better suited for these task types. In contrast, on more complex mathematical reasoning benchmarks like MATH, CHAMP, and WebShop, ToT achieves the highest reasoning accuracy, demonstrating the effectiveness of the multiple-proposal strategy. However,

| Model            | Logical Reasoning |         |         |            |         |         | Web Browsing |         |         | Solving Math Problems |         |         |            |         |         |            |         |         | Commonsense Reasoning |         |         |
|------------------|-------------------|---------|---------|------------|---------|---------|--------------|---------|---------|-----------------------|---------|---------|------------|---------|---------|------------|---------|---------|-----------------------|---------|---------|
|                  | P3                |         |         | SCAN       |         |         | WebShop      |         |         | MATH                  |         |         | CHAMP      |         |         | DROP       |         |         | CSQA                  |         |         |
|                  | Acc               | CTime   | C_API   | Acc        | CTime   | C_API   | Acc          | CTime   | C_API   | Acc                   | CTime   | C_API   | Acc        | CTime   | C_API   | Acc        | CTime   | C_API   | Acc                   | CTime   | C_API   |
| COT (GPT-4o)     | <b>42%</b>        | 35.8    | 4.45€   | <b>68%</b> | 9.21    | 2.75€   | 35%          | 30.9    | 10.65€  | 51.5%                 | 34.5    | 5.34€   | 55.5%      | 26.4    | 4.45€   | 80%        | 11.6    | 1.30€   | 80%                   | 17.0    | 3.60€   |
| TOT (GPT-4o)     | 38%               | 93.1    | 14.55€  | 52%        | 32.5    | 9.82€   | <b>36%</b>   | 62.4    | 47.34€  | <b>63%</b>            | 60.5    | 9.97€   | 57%        | 64.2    | 11.65€  | 80.5%      | 40.2    | 5.41€   | 82%                   | 98.8    | 20.50€  |
| COT (Llama 3-8B) | 5.5%              | 18.1    | N/A     | 17%        | 5.0     | N/A     | 0.0%         | 10.5    | N/A     | 10%                   | 21.1    | N/A     | 19%        | 13.1    | N/A     | 72%        | 3.8     | N/A     | 70%                   | 8.4     | N/A     |
| TOT (Llama 3-8B) | 5.5%              | 58.3    | N/A     | 13%        | 21.8    | N/A     | 1.4%         | 22.5    | N/A     | 29.5%                 | 49.0    | N/A     | 25%        | 68.1    | N/A     | 65%        | 27.8    | N/A     | 68.5%                 | 89.4    | N/A     |
| DataShunt        | 14%               | 25.1    | 2.45€   | 23.5%      | 7.6     | 1.72€   | 34%          | 30.9    | 8.35€   | 16%                   | 24.9    | 1.66€   | 34%        | 19.1    | 2.98€   | 74%        | 8.6     | 0.60€   | 73%                   | 10.4    | 1.28€   |
| DoT (ours)       | 41%               | 23.5    | 1.58€   | 63%        | 5.5     | 1.20€   | 31%          | 17.2    | 4.97€   | 59%                   | 22.6    | 1.02€   | <b>58%</b> | 16.1    | 0.84€   | <b>85%</b> | 4.9     | 0.32€   | <b>82%</b>            | 9.9     | 0.49€   |
| Improvement      | ↓2.38%            | ↓34.36% | ↓64.49% | ↓7.35%     | ↓40.28% | ↓56.36% | ↓13.89%      | ↓72.43% | ↓89.95% | ↓6.35%                | ↓62.72% | ↓89.50% | ↑1.75%     | ↓74.92% | ↓92.79% | ↑5.59%     | ↓87.81% | ↓94.09% | 0%                    | ↓89.98% | ↓97.60% |

表 1: DoT 与基线在 7 个基准测试上的表现。CTime 和 C\_API 是每项任务的平均开销，其中时间消耗以秒为单位，API 成本以美分 (¢) 计量。N/A 出现在仅使用 LLaMA 进行推理而未调用 OpenAI API 密钥的实验中。在每个基准测试中，最高的推理准确率用加粗显示。基线中具有最高 Acc 的结果用下划线标出，将用于计算最后一行的 “提升”。

从每个基准中随机抽取 200 个问题来构成我们的测试集。这些基准分为四类：

- **逻辑推理。** 我们选择 P3 [30] 和 SCAN[20] 进行评估。这些基准更强调逻辑，涉及诸如遍历、逆向推理和异常检测等挑战，这些都需要更高水平的逻辑连贯性和严谨性。
- **数学问题解决。** 这一类别的 benchmark 标记 primarily 涉及数学问题，包括计算、数学知识和问题解决技巧。它被广泛认为是对大型语言模型推理能力的关键测试。我们为这一类别选择了三个基准测试：MATH [14]、CHAMP [25], 和 DROP [9]。
- **常识推理。** CSQA [36] 是一个广泛使用的常识推理数据集，它对推理能力的要求较低，但需要更广泛的知识库。大型语言模型（LLM）参数规模的差异导致它们所拥有的知识系统存在差异，使得 CSQA 非常适合用于评估我们的协作推理场景。

- **网页浏览。** 我们选择 WebShop [42]，它测试 LLM 作为现实场景中的交互代理的能力。它挑战 LLM 代理在网页上导航、理解复杂查询，并采取适当的行动以满足用户请求。该基准测试与 AI 助手场景高度契合，包括指令理解、查询重构和战略性信息探索。

**基线与提示方法。** 鉴于边缘 - 云协作环境的限制，我们建立了三个基线。

- **CoT [38]。** 这个基线将问题分解为子步骤，逐步解决每个子步骤。我们在整个过程中使用单一的 LLM，每个步骤只执行一次，不进行迭代。
- **ToT [43]。** ToT 为每个子步骤探索多种解决方案（M=5）。我们使用评分机制保留得分最高的 N=3 条路径，其中得分最高的路径决定最终答案。ToT 还使用单个大型语言模型进行推理。
- **DataShunt [3]。** 这种方法在每个任务开始时动态选择本地 SLM 和基于云的 LLM。它提示 LLM 评估任务复杂性，从而分配一个解决模型，该模型随后在所有子步骤中一致使用。

在 CoT 和 ToT 中涉及的任务分解过程中，我们采用了在 DoT 中使用的相同提示方法，应用

每个基准有八个手工制作的示例。这些示例是从数据集中随机抽取的，并且与测试集正交。具体的提示可以在附录 B.5 中找到。

**大型语言模型的选择与部署** 我们使用 GPT-4o [29] 作为基于云的 LLM，并使用 Llama 3-8B [27] 作为设备端 LLM，因为其参数规模在理论上可部署于边缘设备。Llama 3-8B 模型部署在本地 A100 GPU 上，上下文长度限制为 8192。我们的适配器参数数量仅为 13,109,249，大约是本地 Llama 参数的 1/800。

**评估。** 对于评估，我们建立了三个指标： *Acc*, *Ctime*, *C\_API*。六个基准测试（不包括 WebShop）的结果是确定性的。例如，CSQA 的结果是单字母选项，可以直接判断正确性。类似地，对于 P3，推理输出可以传入问题函数，如果函数返回 True，则认为推理是正确的。对于 WebShop，我们使用原始基准提供的评估环境，根据约束条件平均所有购买商品的满意度奖励。

## 5.2 主要结果

我们的方法与基线的比较显示在表 1 中，我们用加粗标出每个基准测试中五个基线实验的最高准确率结果，同时将相关成本用下划线标出。我们计算了我们结果相对于准确率最高的基线的相对提升。

实验结果表明，我们的方法在显著降低成本的同时，将准确度保持在可接受的下降范围内。在七个基准测试中，与最佳基线结果相比，准确度的相对变化分别为： -2.38%、 -7.35%、 -13.89%、 -6.35%、 1.75%、 5.59% 和 0%。值得注意的是，在某些情况下观察到了正向提升，这表明我们的框架有潜力增强大型语言模型的推理能力。这一增强在消融研究 5.4 中有进一步讨论。同时，与具有最高准确度的基线相比，我们的方法实现了成本的大幅降低，平均时间减少 66.12 %，平均 API 成本减少 83.57%，远远超过了准确度的下降。

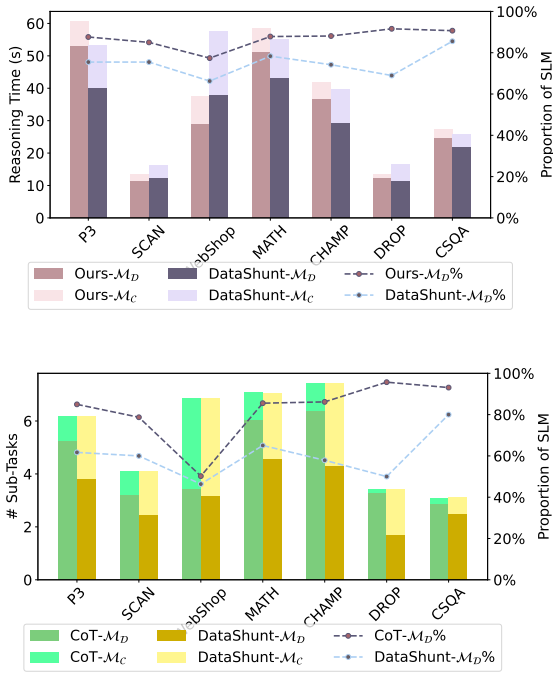
此外，在 P3 和 SCAN 基准测试中，CoT 达到了最佳性能，这表明顺序线性推理结构更适合这些任务类型。相反，在更复杂的数学推理基准测试如 MATH、CHAMP 和 WebShop 中，ToT 达到了最高的推理准确率，展示了多提议策略的有效性。然而，

| Methods              | SLM Ratio | SR     | # Evaluation | API Cost |
|----------------------|-----------|--------|--------------|----------|
| Zero-Shot LLM        | 53.11%    | 92.78% | 1            | \$2.56   |
| Binary-Search        | 69.46%    | 88.45% | 8.456        | \$16.45  |
| $\alpha$ -tree (n=1) | 85.53%    | 99.44% | 3.4589       | \$7.34   |
| $\alpha$ -tree (n=2) | 86.45%    | 96.34% | 2.3445       | \$5.12   |

**Table 2: Methods for Searching Optimal Allocation Scheme**

our approach incurs less than half of CoT’s cost and only one-tenth of ToT’s cost while still achieving comparable performance, highlighting the broad applicability of our graph-structured reasoning and the effectiveness of the edge-cloud collaboration.

To evaluate whether our collaborative approach fully leverages locally deployed SLMs for reasoning, we calculate the proportion of reasoning time by the SLMs relative to the overall reasoning time cost, as well as the percentage of sub-tasks assigned to SLMs. These metrics were compared against the baseline *DataShunt*, and the results are presented in Figure 5. The stacked bar chart illustrates the specific values of SLM and LLM, while the line graph reflects the proportion of SLM. As shown in the figure, across all benchmarks, our allocation strategy makes more comprehensive use of SLMs, with the reasoning time and task allocation percentages exceeding those of the baseline by an average of 11.99% and 21.92%, respectively.

**Figure 5: Proportion of SLMs in time cost and # sub-tasks.**

### 5.3 Efficiency & Quality of Dataset Construction

To validate the effectiveness of our method for difficulty assessment and optimal allocation search in dataset construction, we compared our  $\alpha$ -tree with other baseline search methods. We establish two baseline methods: **Zero-shot LLM**: This approach utilizes a LLM to assess the difficulty of sub-tasks. The evaluation model employs GPT-4o, utilizing a question-and-answer format that incorporates the task content, all sub-tasks, and the current sub-task into the

prompt. **Binary-Search**: This method employs a binary search strategy. Initially, all sub-tasks are processed uniformly using GPT-4o. If the reasoning results are correct, half of the sub-tasks are randomly assigned to Llama 3-8B, with this random assignment repeated N times until successful. In the second round of binary search, half of the remaining sub-tasks assigned to the larger model are again randomly allocated to the smaller model. This process continues until all sub-tasks are either handled by the smaller model or incorrect reasoning persists. We set N to 5.

For our  $\alpha$ -tree approach, which sequentially searches for optimal solutions according to difficulty, we tested with n=1 and n=2 (where n represents the number of models whose allocation is altered at each search step). The comparison metrics included two quality assessment indicators: *SLM Ratio*: the proportion of the smaller on-device model in the final allocation scheme, and *SR*: success rate of reasoning with the final allocation scheme applied. Additionally, we considered two cost metrics: *# Evaluation*: each evaluation requires a complete reasoning pass over all sub-tasks based on the allocation strategy, and *API Cost*, which refers to the expense incurred when invoking the cloud-based LLM API during dataset construction.

We visualize the quality and cost of the constructed dataset on the MATH benchmark. As shown in Table 2, in terms of construction quality, our  $\alpha$ -tree achieved the highest proportion of small model usage while maintaining the highest reasoning accuracy. This demonstrates that our method finds a more efficient collaborative allocation strategy, ensuring high-quality reasoning while minimizing inference costs. *Zero-shot LLM* resulted in the lowest proportion of SLM usage, indicating that LLMs fail to assess the difficulty of sub-tasks. *Binary-Search* also showed a lower SLM proportion compared to ours. While increasing the number of search attempts (N) could potentially yield better results through random search, this unordered and purely random strategy will incur significant costs. Even with the current 5-time search, the cost is already several times higher than our approach. Further scaling will bring unsustainable cost. Naturally, *LLM-Eval* has the lowest cost, but its construction quality is significantly inferior.

### 5.4 Ablation Study

In this ablation study, we demonstrate the impact of the graph structure and task allocation mechanisms on the efficiency and effectiveness of our DoT model. Upon removing the graph structure, a sequential reasoning framework is used instead. In the absence of task allocation, a single language model or simple referral strategy is adopted. As shown in Table 3, the full DoT model, which includes both components, achieves high accuracy across various tasks while maintaining low time and API costs. When the graph component is removed, both accuracy and efficiency are reduced, indicating the impact of graph structure on enhancing reasoning performance and the effectiveness of graph-based parallel execution.

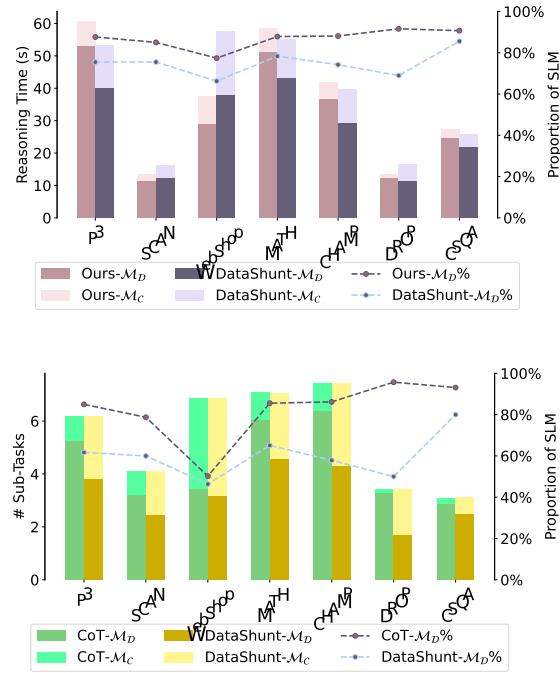
Additionally, removing the task allocation mechanism leads to higher API costs and increased time without substantial accuracy gains. For example, in the P3 task, removing allocation and solely using GPT-4o incurs a cost of 4.80¢ compared to DoT’s 1.58¢, despite only a modest increase in accuracy (43% vs. 41%). This demonstrates that the allocation mechanism helps balance accuracy with efficiency.

| Methods              | SLM Ratio | SR     | # Evaluation | API Cost |
|----------------------|-----------|--------|--------------|----------|
| Zero-Shot LLM        | 53.11%    | 92.78% | 1            | \$2.56   |
| Binary-Search        | 69.46%    | 88.45% | 8.456        | \$16.45  |
| $\alpha$ -tree (n=1) | 85.53%    | 99.44% | 3.4589       | \$7.34   |
| $\alpha$ -tree (n=2) | 86.45%    | 96.34% | 2.3445       | \$5.12   |

**Table 2: Methods for Searching Optimal Allocation Scheme**

我们的方法成本不到 CoT 的一半，且仅为 ToT 成本的十分之一，同时仍能实现可比的性能，突出显示了我们图结构推理的广泛适用性以及边缘 - 云协作的有效性。

为了评估我们的协作方法是否充分利用本地部署的 SLM 进行推理，我们计算了 SLM 推理时间占总体推理时间成本的比例，以及分配给 SLM 的子任务百分比。这些指标与基线 *DataShunt* 进行了比较，结果如图 5 所示。堆叠柱状图展示了 SLM 和 LLM 的具体数值，而折线图反映了 SLM 的比例。如图所示，在所有基准测试中，我们的分配策略更全面地利用了 SLM，推理时间和任务分配百分比分别平均超出基线 11.99% 和 21.92%。

**图 5：SLM 在时间成本和子任务数量中的比例。**

### 5.3 数据集构建的效率与质量

为了验证我们的方法在数据集构建中对任务难度评估和最优分配搜索的有效性，我们将我们的  $\alpha$ -树与其他基线搜索方法进行了比较。我们建立了两种基线方法：**零样本大语言模型（Zero-shot LLM）**：该方法利用大语言模型评估子任务的难度。评估模型使用 GPT-4o，采用问答格式，将任务内容、所有子任务以及当前子任务纳入其中。

提示。**二分查找**：该方法采用二分查找策略。最初，所有子任务都使用 GPT-4o 统一处理。如果推理结果正确，则将一半子任务随机分配给 Llama 3-8B，这种随机分配重复 N 次直到成功。在第二轮二分查找中，分配给较大模型的剩余子任务的一半再次随机分配给较小模型。该过程持续进行，直到所有子任务要么由较小模型处理，要么推理仍然不正确。我们将 N 设为 5。

对于我们的  $\alpha$ -树方法，该方法根据难度顺序地搜索最优解，我们在 n=1 和 n=2（其中 n 表示每个搜索步骤中更改分配的模型数量）下进行了测试。比较指标包括两个质量评估指标：SLM 比例：最终分配方案中较小的设备模型所占的比例，以及 SR：应用最终分配方案后的推理成功率。此外，我们考虑了两个成本指标：# 评估：每次评估都需要根据分配策略对所有子任务进行完整的推理过程，以及 API 成本，指在数据集构建过程中调用基于云的 LLM API 时产生的费用。

我们在 MATH 基准上可视化了构建数据集的质量和成本。如表 2 所示，就构建质量而言，我们的  $\alpha$ -树在保持最高推理准确率的同时，实现了最高比例的小模型使用。这表明我们的方法找到了更高效的协作分配策略，在最小化推理成本的同时确保高质量的推理。零样本 LLM 导致 SLM 使用比例最低，这表明 LLM 无法评估子任务的难度。二分搜索相比于我们的方法，也表现出较低的 SLM 比例。虽然增加搜索次数 (N) 可能通过随机搜索获得更好的结果，但这种无序且纯随机的策略会产生显著成本。即使在当前的 5 次搜索下，成本也已经是我们方法的数倍。进一步扩展将带来不可持续的成本。自然，LLM 评估的成本最低

### 5.4 消融研究

在这项消融研究中，我们演示了图结构和任务分配机制对我们 DoT 模型的效率和有效性的影响。在移除图结构的情况下，使用顺序推理框架代替。在没有任务分配的情况下，采用单一语言模型或简单的推荐策略。如表 3 所示，完整的 DoT 模型包含这两个组件，在各种任务中都能实现高准确性，同时保持较低的时间和 API 成本。当移除图组件时，准确性和效率均有所下降，这表明图结构对增强推理性能以及基于图的并行执行的有效性具有影响。

此外，移除任务分配机制会导致更高的 API 成本和增加的时间，而准确性并没有显著提升。例如，在 P3 任务中，移除分配机制并仅使用 GPT-4o 的成本为 4.80¢，而 DoT 的成本为 1.58¢，尽管准确性只是略微提高（43% 对 41%）。这表明，分配机制有助于在准确性和效率之间取得平衡。

| Model                                | Logical Reasoning |      |       |      |      |       | Web Browsing |      |        | Solving Math Problems |      |       |       |      |       |      |      |       | Commonsense Reasoning |      |       |
|--------------------------------------|-------------------|------|-------|------|------|-------|--------------|------|--------|-----------------------|------|-------|-------|------|-------|------|------|-------|-----------------------|------|-------|
|                                      | P3                |      |       | SCAN |      |       | WebShop      |      |        | MATH                  |      |       | CHAMP |      |       | DROP |      |       | CSQA                  |      |       |
|                                      | Acc               | Time | Api   | Acc  | Time | Api   | Acc          | Time | Api    | Acc                   | Time | Api   | Acc   | Time | Api   | Acc  | Time | Api   | Acc                   | Time | Api   |
| DoT w/o Graph                        | 38%               | 36.4 | 1.44€ | 57%  | 5.2  | 1.18€ | 31%          | 20.7 | 4.97€  | 56%                   | 20.9 | 0.99€ | 52%   | 27.7 | 0.82€ | 80%  | 4.2  | 0.26€ | 77%                   | 9.6  | 0.48€ |
| DoT w/o Allocation (GPT-4o)          | 43%               | 39.8 | 4.80€ | 64%  | 10.7 | 3.66€ | 39.5%        | 19.4 | 13.89€ | 78%                   | 36.9 | 7.68€ | 64%   | 30.4 | 5.05€ | 84%  | 12.8 | 1.69€ | 85%                   | 20.6 | 1.49€ |
| DoT w/o Allocation (Llama 3-8B)      | 29%               | 21.5 | N/A   | 52%  | 4.2  | N/A   | 3.5%         | 10.7 | N/A    | 49%                   | 19.8 | N/A   | 47%   | 15.8 | N/A   | 77%  | 4.3  | N/A   | 76%                   | 8.9  | N/A   |
| DoT w/o Allocation (Simple Referral) | 33%               | 31.1 | 3.21€ | 57%  | 6.3  | 2.41€ | 12.5%        | 13.5 | 6.27€  | 51%                   | 26.3 | 3.54€ | 53.5% | 19.6 | 2.15€ | 80%  | 6.4  | 0.87€ | 79%                   | 14.6 | 0.64€ |
| DoT (ours)                           | 41%               | 23.5 | 1.58€ | 63%  | 5.5  | 1.20€ | 31%          | 17.2 | 4.97€  | 59%                   | 22.6 | 1.02€ | 58%   | 16.1 | 0.84€ | 85%  | 4.9  | 0.32€ | 82%                   | 9.9  | 0.49€ |

Table 3: Result of ablation study.

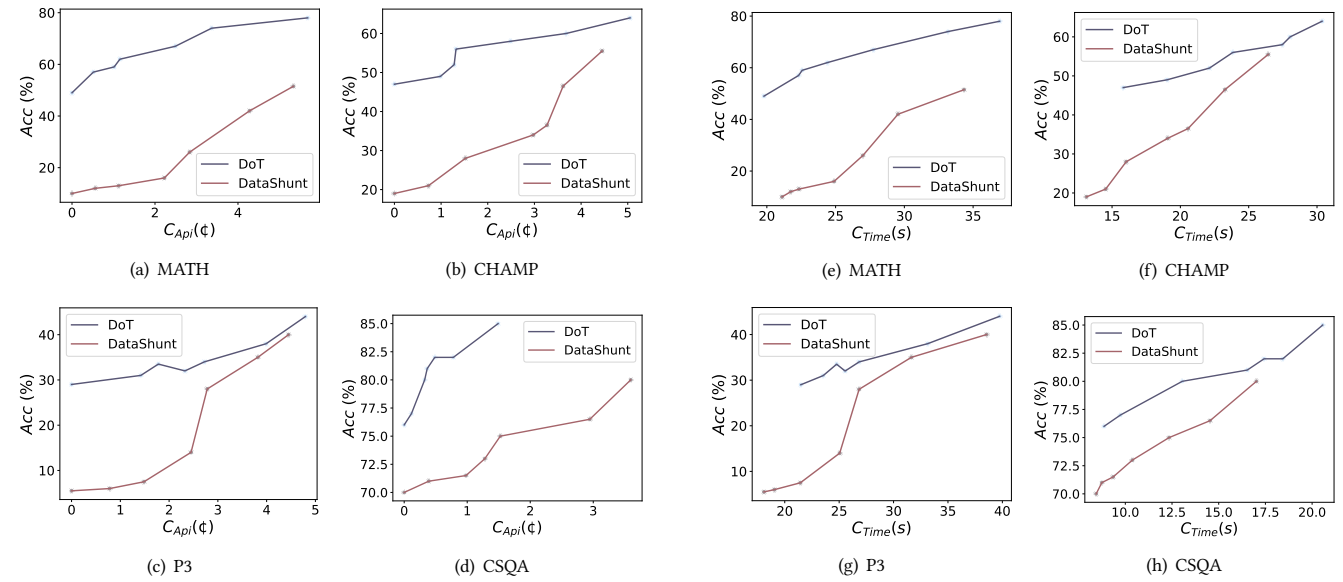
Acc –  $C_{API}$  trade-offAcc –  $C_{Time}$  trade-off

Figure 6: Acc – Cost trade-off curves on 4 benchmarks.

Moreover, We are surprised to discover that, without considering costs, using the cloud-based LLM exclusively can greatly improve the reasoning accuracy on the challenging MATH and CHAMP benchmarks, where accuracy increased from 63% to 78% and from 57% to 64%, respectively, compared to the best baseline.

### 5.5 Trade-off Between Accuracy and Cost

The ablation study demonstrates that our on-graph reasoning method significantly enhances the reasoning capability of LLMs, though at a higher computational cost. In this section, we delve into the detailed relationship between cost and reasoning accuracy to identify the achievable performance upper bound under different budget constraints. Based on our original task allocation strategy, we generate diverse model allocation schemes by proportionally assigning more sub-tasks to either the cloud-based LLM or the local deployed SLM according to the difficulty level of sub-tasks, resulting in the various data points shown in the figure. We also introduce *DataShunt* as a baseline, where the proportion of tasks handled by the LLMs is adjusted by tuning the LLM evaluation thresholds. We test our method on four benchmarks, plotting the trade-off curves for accuracy versus API cost and accuracy versus time cost. As illustrated in Figure 6, our curve consistently remains above the baseline curve. Under the same accuracy, both the token cost and time expenditure

of DoT are significantly lower than those of DataShunt, demonstrating that the model allocation method of DoT is quite efficient. Furthermore, as the budget increases, our DoT approach achieves a higher performance upper bound.

### 6 Conclusion

In conclusion, our proposed *Division-of-Thoughts* collaborative reasoning framework effectively enhances the reasoning performance of LLMs on edge devices by intelligently distributing tasks between local and cloud resources. *DoT* not only breaks down complex tasks into manageable subtasks but also leverages a directed-acyclic graph to optimize the reasoning path, facilitating parallel edge-cloud processing. Furthermore, the plug-and-play adapter accurately assesses the difficulty of sub-tasks for allocation, ensuring high reasoning performance while significantly reducing costs of money and time. Looking ahead, we aim to apply our reasoning framework across a broader and more diverse range of scenarios, striving to contribute to the advancement of edge AI.

### Acknowledgments

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| Model                                | Logical Reasoning |      |       |      |      |       | Web Browsing |      |        | Solving Math Problems |      |       |       |      |       |      |      |       | Commonsense Reasoning |      |       |
|--------------------------------------|-------------------|------|-------|------|------|-------|--------------|------|--------|-----------------------|------|-------|-------|------|-------|------|------|-------|-----------------------|------|-------|
|                                      | P3                |      |       | SCAN |      |       | WebShop      |      |        | MATH                  |      |       | CHAMP |      |       | DROP |      |       | CSQA                  |      |       |
|                                      | Acc               | Time | Api   | Acc  | Time | Api   | Acc          | Time | Api    | Acc                   | Time | Api   | Acc   | Time | Api   | Acc  | Time | Api   | Acc                   | Time | Api   |
| DoT w/o Graph                        | 38%               | 36.4 | 1.44€ | 57%  | 5.2  | 1.18€ | 31%          | 20.7 | 4.97€  | 56%                   | 20.9 | 0.99€ | 52%   | 27.7 | 0.82€ | 80%  | 4.2  | 0.26€ | 77%                   | 9.6  | 0.48€ |
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| DoT w/o Allocation (Llama 3-8B)      | 29%               | 21.5 | N/A   | 52%  | 4.2  | N/A   | 3.5%         | 10.7 | N/A    | 49%                   | 19.8 | N/A   | 47%   | 15.8 | N/A   | 77%  | 4.3  | N/A   | 76%                   | 8.9  | N/A   |
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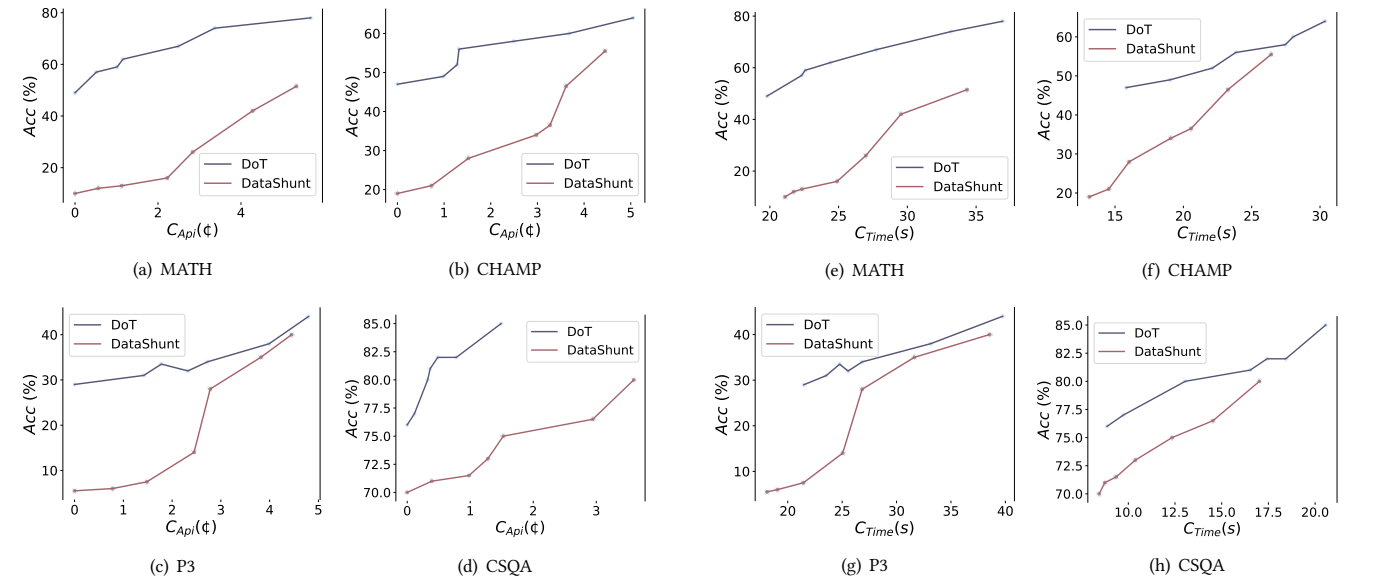
Acc –  $C_{API}$  trade-oAcc –  $C_{Time}$  trade-o

Figure 6: Acc – Cost trade-o curves on 4 benchmarks.

此外，我们惊讶地发现，在不考虑成本的情况下，仅使用基于云的 LLM 可以大大提高在困难的 MATH 和 CHAMP 基准测试上的推理准确性，其中准确率分别从 63% 提升到 78% 以及从 57% 提升到 64%，相比最佳基线。

### 5.5 精度与成本的权衡

消融研究表明，我们的图上推理方法显著增强了大语言模型（LLMs）的推理能力，尽管计算成本更高。在本节中，我们深入探讨成本与推理准确性之间的详细关系，以确定在不同预算约束下可实现的性能上限。基于我们原始的任务分配策略，我们通过根据子任务难度将更多子任务按比例分配给基于云的 LLM 或本地部署的 SLM，生成多样的模型分配方案，从而形成图中所示的各种数据点。我们还引入了 *DataShunt* 作为基线，其中通过调整 LLM 评估阈值来改变 LLM 处理任务的比例。我们在四个基准数据集上测试了我们的方法，绘制了准确率与 API 成本以及准确率与时间成本的权衡曲线。如图 6 所示，我们的方法曲线始终高于基线曲线。

DoT 的结果显著低于 DataShunt，这表明 DoT 的模型分配方法相当高效。此外，随着预算的增加，我们的 DoT 方法能够达到更高的性能上限。

### 6 结论

总之，我们提出的思想分工协作推理框架通过在本地和云端资源之间智能分配任务，有效提升了边缘设备上大模型的推理性能。*DoT* 不仅将复杂任务拆分为可管理的子任务，还利用有向无环图优化推理路径，从而实现边缘 - 云的并行处理。此外，即插即用适配器能够准确评估子任务的难度以进行分配，确保高推理性能，同时显著降低时间和金钱成本。展望未来，我们希望在更多、更广泛的场景中应用我们的推理框架，致力于推动边缘人工智能的发展。

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## Appendix

### A Supplementary Experiments

#### A.1 Generalizability of Adapter

We conduct two supplementary experiments to evaluate the adapter’s generalization performance across different tasks. **Intra-category generalization:** Within three mathematical benchmarks, train on two and test on the remaining one. **Cross-category generalization:** Train on six benchmarks and test on the remaining one. We also evaluate the performance of the adapter in the absence of any training. Results are shown in Table 4, demonstrating that intra-category generalization within math-related benchmarks is almost as effective as task-specific training. And although there is a slight performance decrease in cross-category application, it still outperforms no training.

#### A.2 Evaluation of Task Decomposition Quality

For the quality of the task decomposition, we compare our method with a vanilla prompting strategy, focusing on evaluating the *independence* of subtasks and the impact of the decomposition strategy on both the final reasoning accuracy and the reasoning cost. For *independence*, we aim to determine whether the generated subtasks are independent (i.e., without overlapping mathematical computation) and calculate the independence rate for performance ealuation. We first manually annotate the independence of the decomposition results for 50 tasks in the MATH benchmark. Our method achieves 91.3% in independence rate, significantly outperforming the vanilla prompting strategy, which only reaches 74.2%. This demonstrates the high quality of our task decomposition approach.

Subsequently, we employ GPT-4o to evaluate the independence of subtasks generated by both methods and compared its results to manual annotations. The GPT-4o assessments achieve a 93% consistency with human annotations, indicating that GPT-4o can provide annotation quality comparable to that of humans and can be effectively used for large-scale evaluations. Therefore, we utilize GPT-4o to evaluate the independence of task decomposition across the entire test set, with the results presented in Table 5. The results indicate that our task decomposition method achieves an average improvement of 15.7% in subtask independence compared to the baseline, demonstrating a clear advantage. Moreover, this improvement contributes to enhanced reasoning accuracy and reduced reasoning costs.

#### A.3 Impact of the Number of Sub-tasks on Reasoning Efficiency

As problems become more complex and the number of decomposed sub-tasks increases, whether our framework can achieve a significant improvement in reasoning efficiency while maintaining accuracy remains a valuable research question. Therefore, we conduct supplementary experiment, where we compare our method with baseline that *sequentially reason all sub-tasks on SLM* and perform statistical analyses based on varying numbers of sub-tasks. The results in Table 6 demonstrate that our model consistently maintains high reasoning efficiency across tasks with different numbers of sub-task nodes.

Furthermore, as the number of sub-tasks increases, the enhancement in efficiency becomes increasingly pronounced. When the

| (a) Intra-category generalization: MATH  |       |       |      |       |  |
|------------------------------------------|-------|-------|------|-------|--|
| Train                                    | Test  | Acc   | Time | API   |  |
| MATH                                     | MATH  | 59%   | 22.6 | 1.02¢ |  |
| DROP, CHAMP                              | MATH  | 58%   | 26.9 | 1.28¢ |  |
| /                                        | MATH  | 32%   | 33.2 | 2.18¢ |  |
| (b) Intra-category generalization: CHAMP |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| CHAMP                                    | CHAMP | 58%   | 16.1 | 0.84¢ |  |
| MATH, DROP                               | CHANP | 56.5% | 20.2 | 1.32¢ |  |
| /                                        | CHANP | 27%   | 28.4 | 2.46¢ |  |
| (c) Intra-category generalization: DROP  |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| DROP                                     | DROP  | 85%   | 4.9  | 0.32¢ |  |
| MATH, CHAMP                              | DROP  | 81.0% | 6.4  | 0.43¢ |  |
| /                                        | DROP  | 72.0% | 7.2  | 0.49¢ |  |
| (d) Cross-category generalization: CSQA  |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| CSQA                                     | CSQA  | 82%   | 9.9  | 0.49¢ |  |
| DROP,CHAMP,MATH,P3,SCAN,Webshop          | CSQA  | 76%   | 12.5 | 0.58¢ |  |
| /                                        | CSQA  | 70%   | 15.2 | 0.84¢ |  |
| (e) Cross-category generalization: MATH  |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| MATH                                     | MATH  | 59%   | 22.6 | 1.02¢ |  |
| DROP,CHAMP,P3,SCAN,Webshop,CSQA          | MATH  | 53%   | 29.1 | 1.34¢ |  |
| /                                        | MATH  | 32%   | 33.2 | 2.18¢ |  |
| (f) Cross-category generalization: CHAMP |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| CHAMP                                    | CHAMP | 58%   | 16.1 | 0.84¢ |  |
| DROP,MATH,P3,SCAN,Webshop,CSQA           | CHAMP | 50%   | 19.6 | 1.35¢ |  |
| /                                        | CHAMP | 27%   | 28.4 | 2.46¢ |  |
| (g) Cross-category generalization: P3    |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| P3                                       | P3    | 41%   | 23.5 | 1.58¢ |  |
| DROP,CHAMP,MATH,SCAN,Webshop,CSQA        | P3    | 33%   | 29.3 | 1.73¢ |  |
| /                                        | P3    | 15%   | 35.1 | 2.38¢ |  |

Table 4: Evaluation of Generalizability

number of subtasks exceeds eight, the reduction in reasoning time surpasses even 30%. This outcome is inherently logical, for as the quantity of sub-tasks escalates, the task graph grows in complexity, thereby expanding the potential for parallel reasoning and amplifying the advantages in terms of reasoning time.

## B Benchmarks and Implementation Details

### B.1 Mathematics

- **MATH** [14]. Mathematics Aptitude Test of Heuristics (MATH), comprises 12,500 problems from prestigious U.S. mathematics competitions like AMC and AIME. These problems, collected from platforms such as AoPS, test advanced problem-solving skills beyond standard K-12 math. Each problem

## 附录

### 补充实验

#### A.1 适配器的普适性

我们进行了两个补充实验，以评估适配器在不同任务中的泛化性能。 **类别内泛化**：在三个数学基准内，训练两个基准并测试剩下的一个。 **跨类别泛化**：在六个基准上训练，并测试剩下的一个。我们还评估了适配器在没有任何训练情况下的性能。结果如表 4 所示，表明在数学相关基准内的类别内泛化几乎与针对特定任务的训练一样有效。虽然跨类别应用中性能略有下降，但仍然优于没有训练的情况。

#### A.2 任务分解质量评估

为了评估任务分解的质量，我们将我们的方法与普通提示策略进行比较，重点评估子任务的独立性以及分解策略对最终推理准确性和推理成本的影响。对于独立性，我们的目标是确定生成的子任务是否独立（即没有重叠的数学计算），并计算独立率以评估性能。我们首先人工标注了 MATH 基准中 50 个任务分解结果的独立性。我们的方法在独立率上达到了 91.3%，显著优于仅为 74.2% 的普通提示策略。这证明了我们任务分解方法的高质量。

随后，我们使用 GPT-4o 来评估两种方法生成的子任务的独立性，并将其结果与人工标注进行比较。GPT-4o 的评估与人工标注达到 93% 的一致性，这表明 GPT-4o 可以提供与人工相当的标注质量，并且可以有效用于大规模评估。因此，我们利用 GPT-4o 对整个测试集的任务拆解独立性进行评估，结果如表 5 所示。结果表明，我们的任务拆解方法在子任务独立性方面比基线方法平均提高了 15.7%，显示出明显优势。此外，这一改进有助于提升推理准确性并降低推理成本。

#### A.3 子任务数量对推理效率的影响

随着问题变得更加复杂以及分解子任务的数量增加，我们的框架是否能够在保持准确性的同时实现推理效率的显著提升，仍然是一个有价值的研究问题。因此，我们进行了补充实验，在实验中我们将我们的方法与基线方法进行比较，后者是在 SLM 上依次推理所有子任务，并基于不同数量的子任务进行统计分析。表 6 的结果表明，我们的模型在不同数量的子任务节点的任务中始终保持较高的推理效率。

此外，随着子任务数量的增加，效率的提升变得越来越明显。当

| (a) 类内泛化：MATH rai 测试准确                   |       |       |      |       |  |
|------------------------------------------|-------|-------|------|-------|--|
| Train                                    | Test  | Acc   | Time | API   |  |
| MATH                                     | MATH  | 59%   | 22.6 | 1.02¢ |  |
| DROP, CHAMP                              | MATH  | 58%   | 26.9 | 1.28¢ |  |
| /                                        | MATH  | 32%   | 33.2 | 2.18¢ |  |
| (b) 类内泛化：CHAMP ai 测试准确率                  |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| CHAMP                                    | CHAMP | 58%   | 16.1 | 0.84¢ |  |
| MATH, DROP                               | CHANP | 56.5% | 20.2 | 1.32¢ |  |
| /                                        | CHANP | 27%   | 28.4 | 2.46¢ |  |
| (c) Intra-category generalization: DROP  |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| DROP                                     | DROP  | 85%   | 4.9  | 0.32¢ |  |
| MATH, CHAMP                              | DROP  | 81.0% | 6.4  | 0.43¢ |  |
| /                                        | DROP  | 72.0% | 7.2  | 0.49¢ |  |
| (d) 跨类别泛化：CSQA                           |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| CSQA                                     | CSQA  | 82%   | 9.9  | 0.49¢ |  |
| DROP,CHAMP,MATH,P3,SCAN,Webshop          | CSQA  | 76%   | 12.5 | 0.58¢ |  |
| /                                        | CSQA  | 70%   | 15.2 | 0.84¢ |  |
| (e) Cross-category generalization: MATH  |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| MATH                                     | MATH  | 59%   | 22.6 | 1.02¢ |  |
| DROP,CHAMP,P3,SCAN,Webshop,CSQA          | MATH  | 53%   | 29.1 | 1.34¢ |  |
| /                                        | MATH  | 32%   | 33.2 | 2.18¢ |  |
| (f) Cross-category generalization: CHAMP |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| CHAMP                                    | CHAMP | 58%   | 16.1 | 0.84¢ |  |
| DROP,MATH,P3,SCAN,Webshop,CSQA           | CHAMP | 50%   | 19.6 | 1.35¢ |  |
| /                                        | CHAMP | 27%   | 28.4 | 2.46¢ |  |
| (g) Cross-category generalization: P3    |       |       |      |       |  |
| Train                                    | Test  | Acc   | Time | API   |  |
| P3                                       | P3    | 41%   | 23.5 | 1.58¢ |  |
| DROP,CHAMP,MATH,SCAN,Webshop,CSQA        | P3    | 33%   | 29.3 | 1.73¢ |  |
| /                                        | P3    | 15%   | 35.1 | 2.38¢ |  |

Table 4: 一般的上的验证能力

子任务数量超过八个时，推理时间的减少甚至超过 30%。这一结果本质上是合乎逻辑的，因为随着子任务数量的增加，任务图的复杂性增加，从而扩大了并行推理的潜力，并增强了推理时间方面的优势。

## B 基准和实现细节

### B.1 数学

- **数学** [14]。启发式数学能力测试（MATH），包括来自美国著名数学竞赛如 AMC 和 AIME 的 12,500 道题目。这些题目收集自 AoPS 等平台，测试超出标准 K-12 数学的高级问题解决能力。每道题

(a) Decomposition Independence Evaluation on CSQA

| Method                          | Independence | Acc | Time | API   |
|---------------------------------|--------------|-----|------|-------|
| (Ours) DoT decomposition        | 95%          | 82% | 9.9  | 0.49¢ |
| vanilla prompting decomposition | 87%          | 81% | 13.2 | 0.77¢ |

(b) Decomposition Independence Evaluation on MATH

| Method                          | Independence | Acc | Time | API   |
|---------------------------------|--------------|-----|------|-------|
| (Ours) DoT decomposition        | 90%          | 59% | 22.6 | 1.02¢ |
| vanilla prompting decomposition | 76%          | 57% | 38.4 | 1.78¢ |

(c) Decomposition Independence Evaluation on P3

| Method                          | Independence | Acc   | Time | API   |
|---------------------------------|--------------|-------|------|-------|
| (Ours) DoT decomposition        | 86%          | 41%   | 23.5 | 1.58¢ |
| vanilla prompting decomposition | 71%          | 36.5% | 24.5 | 2.23¢ |

Table 5: Evaluation of Decomposition Independence

| # Subtask | P3        |            | MATH      |            |
|-----------|-----------|------------|-----------|------------|
|           | ↑Accuracy | ↓Cost_time | ↑Accuracy | ↓Cost_time |
| 3         | 14.5%     | 18.3%      | 9%        | 13.5%      |
| 4         | 18%       | 21.7%      | 13%       | 15.5%      |
| 5         | 20%       | 25.4%      | 14%       | 17.2%      |
| 6         | 13%       | 22.6%      | 17%       | 18.4%      |
| 7         | 14%       | 23.9%      | 15%       | 17.6%      |
| 8         | 12%       | 28.7%      | 13%       | 25.7%      |
| >8        | 17.5%     | 32.1%      | 18%       | 30.3%      |

Table 6: Evaluation of DoT’s Scalability

includes a step-by-step solution and a final answer. The dataset spans seven subjects: Prealgebra, Algebra, Number Theory, Counting and Probability, Geometry, Intermediate Algebra, and Precalculus, with difficulty levels ranging from 1 (easy) to 5 (challenging), allowing models to learn and apply various mathematical heuristics.

- **CHAMP** [25]. Concept and Hint-Annotated Math Problems (CHAMP). This benchmark features 270 non-routine, competition-level problems sourced from Engel’s Problem-Solving Strategies. The problems span five categories: number theory, polynomial, sequence, inequality, and combinatorics, requiring creative strategies and specific tricks. Each problem includes a final checkable answer and a step-by-step solution in natural language. The dataset contains 54 concepts and 330 hints, averaging 1.4 concepts, 1.7 hints, and 6 solution steps per problem. Problem statements average 20.2 words, with solutions averaging 10.9 words per step, highlighting the dataset’s complexity and challenge.
- **DROP** [9]. Discrete Reasoning Over Paragraphs (DROP), is an English reading comprehension dataset with 96k adversarially-created questions. It challenges systems to resolve references in a question and perform discrete reasoning operations like addition, counting, or sorting over multiple input positions within a paragraph. The dataset is crowdsourced and derived from Wikipedia passages designed for complex questions. DROP demands a deeper understanding of paragraph content than previous datasets, with rigorous validation to ensure the quality of its development and test sets.

## B.2 Logic

- **P3** [30]. Python Programming Puzzles (P3), introduces a new type of programming challenge for evaluating program synthesis. P3 contains 397 Python-based puzzles, where the goal is to find an input that makes a given function return True. The puzzles span various difficulty levels, from simple string manipulations to complex algorithmic problems.
- **Scan** [20]. This benchmark is used to evaluate the sequence-to-sequence learning ability to translate simplified natural language commands into action sequences. SCAN contains 20,910 commands generated by a phrase-structure grammar, which describe basic actions such as "jump" or "walk" and their combinatorial variations (e.g., "jump around left"). The logical structure and constraints involved in the translation process make SCAN ideal for assessing LLM reasoning capabilities.

## B.3 Commonsense

- **COMMONSENSEQA** [36]. This benchmark is a challenging dataset designed to test commonsense question answering. It consists of 12,247 multiple-choice questions created using concepts from CONCEPTNET. Each question is authored by crowd-workers to differentiate between multiple target concepts that share a semantic relation with a source concept, encouraging the use of prior knowledge and complex reasoning.

## B.4 On-device AI Assistant Application

- **Webshop** [42]. WebShop serves as a simulated e-commerce environment featuring 1.18 million real-world products and 12,087 crowd-sourced text instructions. Designed to evaluate language agents, it challenges them to navigate diverse web pages and perform actions based on natural language product specifications. Agents encounter obstacles such as interpreting compositional instructions, reformulating queries, and understanding noisy text on webpages, while strategically exploring to fulfill product requirements. The modular design of WebShop separates website navigation from task-specific elements, allowing for easy adaptation to new tasks and domains. This dataset provides a robust platform for assessing the capabilities of language agents in an interactive, real-world-inspired setting, emphasizing their ability to comprehend and act on complex instructions.

## B.5 Implementation Details across Benchmarks

**MATH.** For MATH tasks, we employ a structured workflow comprising four key stages: Task Decomposition, Model Allocation, Dependency Graph Construction, and Step-by-Step Reasoning Based on the Graph. In the task decomposition phase, we prompt the LLM with exemplars of manually decomposed complex problems. The LLM is then instructed to generate manageable subtasks that collectively solve the primary challenge. Subsequently, we task the LLM with establishing subtask dependencies, where a relationship  $Step_i \rightarrow Step_j$  indicates that  $Step_i$  must precede  $Step_j$ . Using the derived dependencies, we construct a reasoning graph via Breadth-First Search (BFS). Subtasks at the same depth are processed in parallel. Upon completion of all subtasks, we conduct a final query

(a) CSQA 方法上的分解独立性评估 I 依赖性准确率 T

| M                               | n   | n | n   |      | ime   | API |
|---------------------------------|-----|---|-----|------|-------|-----|
| (Ours) DoT decomposition        | 95% |   | 82% | 9.9  | 0.49¢ |     |
| vanilla prompting decomposition | 87% |   | 81% | 13.2 | 0.77¢ |     |

(b) 在 MATH 上的分解独立性评估

| Method                          | Independence | Acc | Time | API   |
|---------------------------------|--------------|-----|------|-------|
| (Ours) DoT decomposition        | 90%          | 59% | 22.6 | 1.02¢ |
| vanilla prompting decomposition | 76%          | 57% | 38.4 | 1.78¢ |

(c) P3 方法 I 依赖性累积的分解独立性评估

| M                               | n   | n | n     |      | Time  | API |
|---------------------------------|-----|---|-------|------|-------|-----|
| (Ours) DoT decomposition        | 86% |   | 41%   | 23.5 | 1.58¢ |     |
| vanilla prompting decomposition | 71% |   | 36.5% | 24.5 | 2.23¢ |     |

表 5：分解独立性评估

| # Subtask | P3        |            | MATH      |            |
|-----------|-----------|------------|-----------|------------|
|           | ↑Accuracy | ↓Cost_time | ↑Accuracy | ↓Cost_time |
| 3         | 14.5%     | 18.3%      | 9%        | 13.5%      |
| 4         | 18%       | 21.7%      | 13%       | 15.5%      |
| 5         | 20%       | 25.4%      | 14%       | 17.2%      |
| 6         | 13%       | 22.6%      | 17%       | 18.4%      |
| 7         | 14%       | 23.9%      | 15%       | 17.6%      |
| 8         | 12%       | 28.7%      | 13%       | 25.7%      |
| >8        | 17.5%     | 32.1%      | 18%       | 30.3%      |

Table 6: Evaluation of DoT’s Scalability

包括逐步的解决方案和最终答案。数据集涵盖七个学科：初等代数、代数、数论、计数与概率、几何、中级代数和预备微积分，难度级别从 1（简单）到 5（具有挑战性）不等，使模型能够学习和应用各种数学启发式方法。

- **CHAMP** [25]。概念与提示注释数学问题（CHAMP）。该基准包含 270 道非常规、竞赛水平的问题，来源于恩格尔的《问题解决策略》。这些问题涵盖五类：数论、多项式、序列、不等式和组合学，要求运用创造性策略和特定技巧。每道题都包含可最终检查的答案以及自然语言的逐步解答。数据集包含 54 个概念和 330 条提示，每道题平均有 1.4 个概念、1.7 条提示和 6 个解题步骤。题目陈述平均为 20.2 个词，解答步骤平均为每步 10.9 个词，突显了该数据集的复杂性和挑战性。

- **DROP** [9]。段落离散推理（DROP）是一个英文阅读理解数据集，包含 96,000 个对抗性创建的问题。它要求系统在问题中解析引用，并执行离散推理操作，如加法、计数或对段落中多个输入位置进行排序。该数据集由众包创建，并衍生自为复杂问题设计的维基百科段落。与以往的数据集相比，DROP 需要对段落内容有更深入的理解，并通过严格的验证来确保其开发集和测试集的质量。

## B.2 逻辑

- **P3** [30]。Python 编程难题 (P3) 引入了一种用于评估程序合成的新型编程挑战。P3 包含 397 个基于 Python 的难题，目标是找到一个输入，使给定的函数返回 True。这些难题涵盖了各种难度等级，从简单的字符串操作到复杂的算法问题。
- **扫描** [20]。这个基准用于评估序列到序列学习能力，将简化的自然语言命令翻译为动作序列。SCAN 包含由短语结构语法生成的 20,910 条命令，这些命令描述了基本动作，例如“跳跃”或“行走”，以及它们的组合变化（例如，“向左跳动”）。翻译过程中涉及的逻辑结构和约束使 SCAN 成为评估大型语言模型推理能力的理想工具。

## B.3 常识

- **COMMONSENSEQA** [36]。这个基准是一个具有挑战性的数据集，旨在测试常识问答能力。它由 12,247 个多项选择题组成，这些题目是利用 CONCEPTNET 的概念创建的。每个问题都是由众包工作人员撰写的，目的是在多个与源概念有语义关系的目标概念之间进行区分，从而鼓励使用先验知识和复杂推理。

## B.4 设备端 AI 助手应用

- **网络商店** [42]。WebShop 作为一个模拟的电子商务环境，提供了 118 万种真实世界的产品和 12,087 条众包文本指令。它旨在评估语言代理，挑战代理在不同网页上导航并根据自然语言产品规格执行操作的能力。代理会遇到各种障碍，例如解释组合指令、重新制定查询、理解网页上的噪声文本，同时还需策略性地探索以满足产品需求。WebShop 的模块化设计将网站导航与特定任务元素分离，使其能够轻松适应新的任务和领域。该数据集提供了一个强有力的平台，用于评估语言代理在互动的、受真实世界启发的环境中的能力，强调其理解和执行复杂指令的能力。

## B.5 各基准的实现细节

**数学。**对于数学任务，我们采用一个结构化的工作流程，包括四个关键阶段：任务分解、模型分配、依赖图构建，以及基于图的逐步推理。在任务分解阶段，我们通过手动分解的复杂问题示例来提示大语言模型（LLM）。然后指示 LLM 生成可管理的子任务，这些子任务共同解决主要问题。随后，我们让 LLM 建立子任务之间的依赖关系，其中关系表示  $Step_i$  必须在  $Step_j$  之前完成。利用得出的依赖关系，我们通过广度优先搜索（BFS）构建推理图。具有相同深度的子任务会并行处理。在所有子任务完成后，我们进行最终查询。

to obtain the ultimate solution. This solution undergoes LLM-based evaluation through comparison with the ground truth.

**CHAMP.** For CHAMP tasks, the process is largely similar to the MATH process with the primary distinction being the specific few-shot examples of human-written decompositions provided.

**DROP.** For DROP tasks, we adapt our approach to accommodate the format of questions based on given texts. The prompt for each step incorporates relevant background information provided in the dataset while maintaining the core implementation structure used in other benchmarks.

**P3.** The Programming Puzzle tasks present a unique challenge, requiring the LLM to generate inputs that yield a 'True' output for a given function. In addition to the puzzle description, we provide the LLM with the expected data type of the final input. The evaluation process for P3 differs from other benchmarks: we execute the program using the LLM-generated input and assess the correctness based on the program's output.

**SCAN.** For SCAN tasks focused on translating natural language instructions into action sequences, the steps until evaluation stay the same. The final phase involves converting the model's natural language outputs into standardized action sequences using few-shot examples. The evaluation is conducted by directly comparing these sequences with the true answers, ensuring the outputs accurately match the expected actions.

**COMMONSENSEQA.** For CSQA tasks, presented as multiple-choice questions based on common sense, our implementation has an identical structure as other tasks, while both the problem and its options are presented to the LLM for the following task decomposition and reasoning. The Final Evaluation consists of the LLM choosing the most plausible answer from the provided options. This selected answer is cleaned and will directly compare with the correct choice.

**WebShop.** The WebShop task presents unique challenges due to its interactive nature, where each action influences subsequent states and available options. Unlike our other experiments, WebShop is not perfect for the construction of a complete, predefined reasoning graph. Instead, our framework dynamically generates and executes sub-tasks based on the current state of the shopping session, in which the shopping process is divided into high-level sub-tasks/thoughts. We utilized the Model Allocation for each step when generating an action, either be 'think[]', 'click[]', or 'plan[]'. The first step would be to generate a keyword from the instructions to search on the website. Then, the top  $N = 10$  matched items are recorded for the following procedures. With the conversation from the webpage, we implemented the task decomposition process to get a roadmap for the following actions. Prompting each step at once to the LLMs, the detailed information about each item will be recorded in the search history. Then, an evaluation and comparison process is prompted to choose the best item from the list. Lastly, the LLM will be prompted dynamically based on the previous steps, and self-navigated to complete the final purchase. This implementation demonstrates our framework's flexibility in handling tasks with dynamic, state-dependent decision-making processes.

## C Prompts

### C.1 Task Decomposition

I will now give you a [Based on the type of problem]. The type of problem is type. Please break this problem down into several easy-to-solve steps.

1 examples are as follows: [Manual Written Examples]  
Now the command is question, please decompose it into easy-to-solve steps like the examples. Answer Format: (Please write each broken-down question step on a separate line, starting with a number.)

To solve the question "xxx", we need to know: "1. question  $step_1$ ", "2. question  $step_2$ ", "3. question  $step_3$ ". ...

### C.2 Dependency Construction

*System Prompt:*

Now we have a problem, which we have broken down into many sub-problems. I want you to understand the connection between these sub-problems

*User Prompt:*

The init problem is question. And the sub-problems are steps. Please provide your understanding of the relationships between these sub-problems. Your response must be concise.

Now we need to create standardized connections for the relationships between these sub-problems. Now Given the following subtasks for question: [question], determine the dependencies between them:

[List of Steps]  
Please list the dependencies in the format 'Subproblem A [xxx] -> Subproblem B [xxx]' indicating that Sub-problem A must be completed before Sub-problem B can start. Please identify any potential conditional dependencies from a logical perspective.

Answer format: (Please strictly follow the format. Each dependency should be separated by a new line. No explanation is required.)

Step  $ID_i$  [ sub-problem i ] -> Step  $ID_j$  [ sub-problem j ]

Step  $ID_j$  [ sub-problem m ] -> Step  $ID_n$  [ sub-problem n ] ...

### C.3 Subtask Reasoning

*System Prompt:*

Here is a math word problem. I will first provide a passage of the problem to set the context. Then, I will ask a specific question that requires you to use the information from the problem description, along with calculation and reasoning, to solve it.

Passage: [passage] Question: [question]

I have broken this math question down into several smaller questions. I will assign you sub-questions one by one, and provide the results of previous sub-questions as a reference for your reasoning. Please solve the question according to mathematical logic.

*For each steps* So far, the answers to the resolved sub-questions are as follows: The format is Sub-question-Id: xxx; Sub-question: xxx; Answer: xxx. Sub-question-Id: [Corresponding ID]; Sub-question: [Corresponding Step]; Answer: [Corresponding Solution for the step]

Among them, sub-questions predecessors are directly related to this sub-question, so please pay special attention to them. The sub-question to solve now is xxx: {subtask} Based on the information above, please provide a concise and clear answer

以获得最终解决方案。该解决方案通过与实际情况的比较，进行基于大型语言模型（LLM）的评估。

**CHAMP.** 对于 CHAMP 任务，这个过程在很大程度上与 MATH 过程类似，主要区别在于提供了少量人类编写的分解示例。

**DROP.** 对于 DROP 任务，我们调整了方法以适应基于给定文本的问题格式。每一步的提示都结合了数据集中提供的相关背景信息，同时保持在其他基准测试中使用的核心实现结构。

**P3.** 编程难题任务提出了独特的挑战，要求大型语言模型（LLM）生成能够使给定函数输出“True”的输入。除了难题描述之外，我们还向 LLM 提供了最终输入的预期数据类型。P3 的评估过程不同于其他基准测试：我们使用 LLM 生成的输入执行程序，并根据程序的输出评估正确性。

**扫描。** 对于专注于将自然语言指令翻译成动作序列的 SCAN 任务，直到评估的步骤保持不变。最后阶段包括使用少量示例将模型的自然语言输出转换为标准化的动作序列。评估通过将这些序列与真实答案直接比较进行，以确保输出准确匹配预期的动作。

**COMMONSENSEQA.** 对于 CSQA 任务，其形式为基于常识的多项选择题，我们的实现结构与其他任务相同，同时将问题及其选项提供给大型语言模型（LLM）以进行后续的任务分解和推理。最终评估由 LLM 从提供的选项中选择最合理的答案。所选答案将进行清理，并与正确答案直接比较。

**网络商店（WebShop）。** 由于其互动性，WebShop 任务呈现出独特的挑战，每一个操作都会影响后续的状态和可用选项。与我们的其他实验不同，WebShop 并不适合构建完整的、预先定义的推理图。相反，我们的框架根据购物会话的当前状态动态生成并执行子任务，在其中购物过程被划分为高层次的子任务 / 思路。我们在生成动作时为每一步使用了模型分配，可以是‘思考（think[]）’，‘点击（click[]）’或‘计划（plan[]）’。第一步是根据指令生成一个在网站上搜索的关键词。然后，将排名前  $N = 10$  的匹配商品记录下来以用于后续流程。在网页对话的基础上，我们实施了任务分解过程，以获得后续行动的路线图。每一步同时提示给大型语言模型（LLMs），以获取详细信息关于

## C 提示

### C.1 任务分解

我现在将给你一个 [ 基于问题类型 ] 的问题。问题的类型是 type。请将此问题分解为几个易于解决的步骤。

1 个例子如下：[ 手写示例 ]  
现在命令是一个问题，请像例子一样将其分解为易于解决的步骤。回答格式：（请将每个分解的问题步骤分别写在一行上，以数字开头。）

要解决问题“xxx”，我们需要知道：1. 问题  $step_1$ ，2. 问题  $step_2$ ，3. 问题  $step_3$ 。 ...

### C.2 依存结构

系统提示：  
现在我们有一个问题，我们已经将其分解成许多子问题。我希望你理解这些子问题之间的联系

用户提示：  
初始化问题是主要问题。子问题是步骤。请提供你对这些子问题之间关系的理解。你的回答必须简明扼要。

现在我们需要为这些子问题之间的关系创建标准化连接。现在，给定问题的以下子任务：[ 问题 ]，确定它们之间的依赖关系：

[ 步骤列表 ]  
请以‘子问题 A [xxx]-> 子问题 B [xxx]’的格式列出依赖关系，表示子问题 A 必须在子问题 B 开始之前完成。请从逻辑的角度识别任何潜在的条件依赖关系。

答案格式：（请严格遵循此格式。每个依赖项应换行分隔。无需解释。）步骤  $ID_i$  [ 子问题 i ] -> 步骤  $ID_j$  [ 子问题 j ] 步骤  $ID_j$  [ 子问题 m ] -> 步骤  $ID_n$  [ 子问题 n ] ...

### C.3 子任务推理

系统提示：  
这里有一个数学应用题。我将先提供题目的段落来设置背景。然后，我会提出一个具体问题，需要你使用题目描述中的信息，结合计算和推理来解决它。  
段落：[ 段落 ] 问题：[ 问题 ]  
我已经将这个数学题分解成几个较小的问题。我会一个接一个地分配子问题，并提供前一个子问题的结果作为你推理的参考。请根据数学逻辑来解题。

对于每一步到目前为止，已解决的子问题的答案如下：格式为 子问题编号 : xxx; 子问题 : xxx; 答案 : xxx。子问题编号 : [ 对应的 ID]; 子问题 : [ 对应的步骤 ]; 答案 : [ 对应该步骤的解决方案 ]  
其中，子问题的前置问题与此子问题直接相关，因此请特别关注它们。目前需要解决的子问题是 xxx: {subtask} 基于上述信息，请提供简明清晰的答案